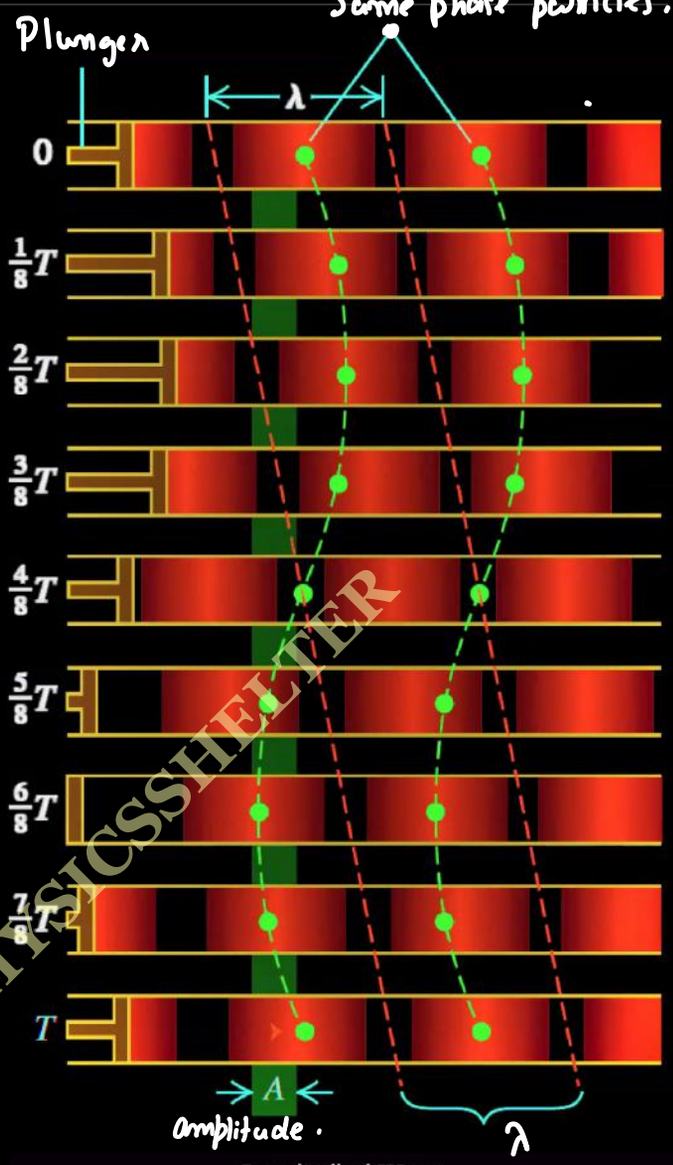
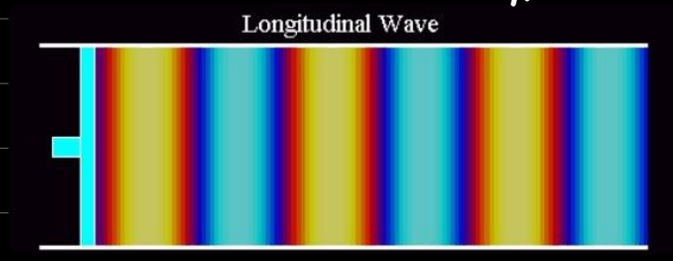
Generation Of Longitudinal Waves:

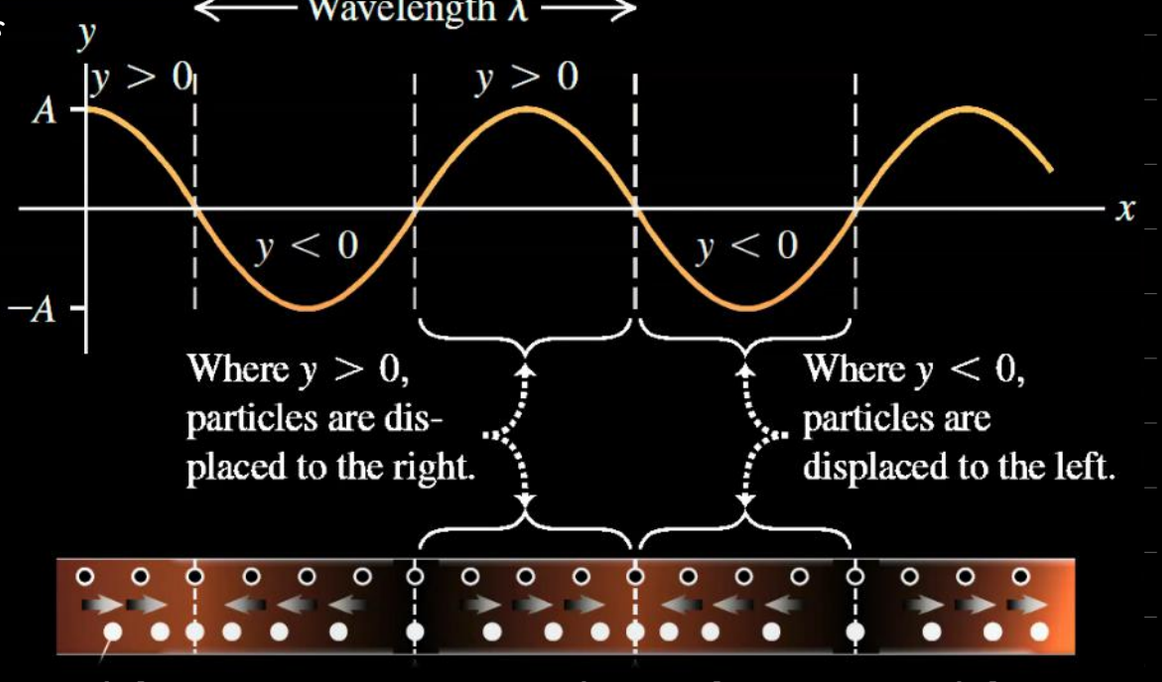
Longitudinal waves can be generated in fluids and solids. The longitudinal waves involve to and fro motion of particles of the medium in the direction of propagation of disturbance. Sound waves are an example of longitudinal waves. Longitudinal waves may also propagate in 2D or 3D however for purpose of JEE we will mainly stick to one dimensional propagation. One way of creating such a wave is to move a piston to and fro inside a cylinder containing air. If the piston is moved simple harmonically, it will create regions of compressions and rarefactions which move along the pipe even though particles just oscillate about their mean positions.





Plotting A Longitudinal Wave:

A longitudinal wave is plotted just like transverse wave. Coordinates on y axis represent particle displacements while on x-axis, it represents natural coordinate.



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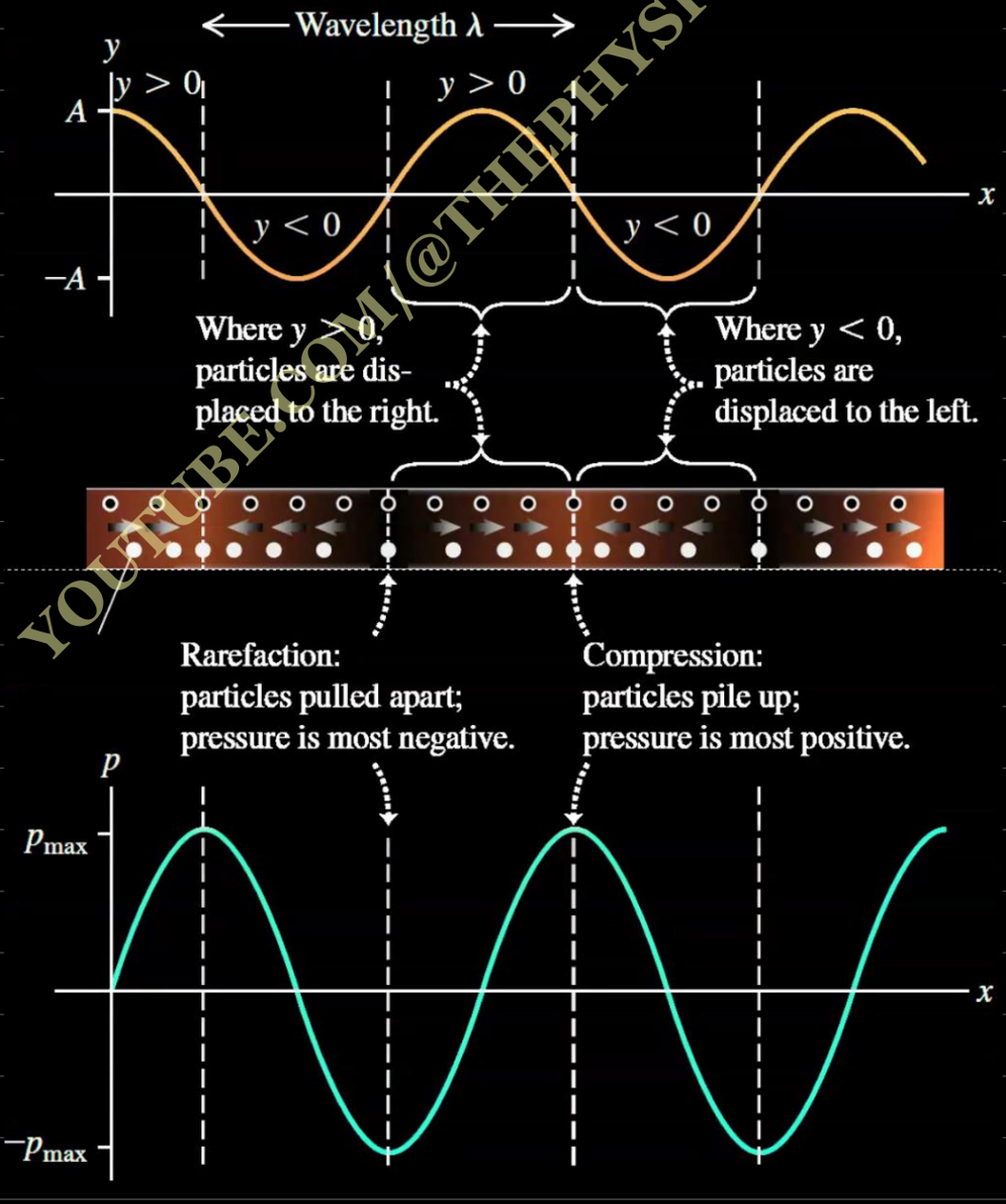
Think Yourself: Where are the regions of compression and rarefaction? Can you relate it to slope of the displacement graph?

Solution: Imagine a row of students equidistant from each other. Now if each student moves forward, however one to the right moves less forward than the one to the left, then their spacing would decrease. Thus if rate of change of displacement wrt to natural position is negative, it creates a compression. (Which means a negative slope of displacement graph would correspond to a compression). Similarly you can figure out that positive slope of displacement vs x graph represents a rarefaction. The regions of compression correspond to higher gauge pressures and the regions of rarefaction lead to lower gauge pressures. This leads to following situation:



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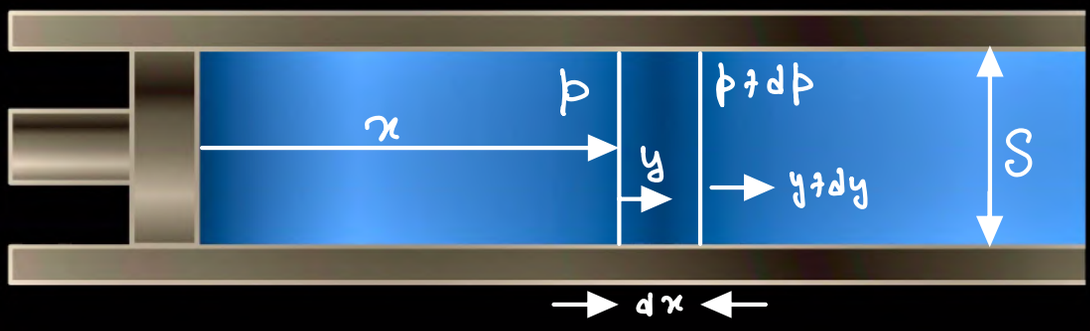
Relation Between Displacment and Strain Wave:

p: excess pressure
x: natural coordinate
y: displacement
S: Cross Sectional Area.



Consider a column of gas as shown in the figure above. We consider FBD of a layer of natural thickness dx. Its left end displaces by y and the right end displaces by y+dy. Now let us calculate volumetric strain in this volume. For this volume element:

Natural Volume: $V = S\,dx$, Deformed Volume: $V' = S(x+dx+y+dy-(x+y))$

$\Rightarrow$ Deformed Volume: $V' = S(dx+dy)$, thus volume deformation:

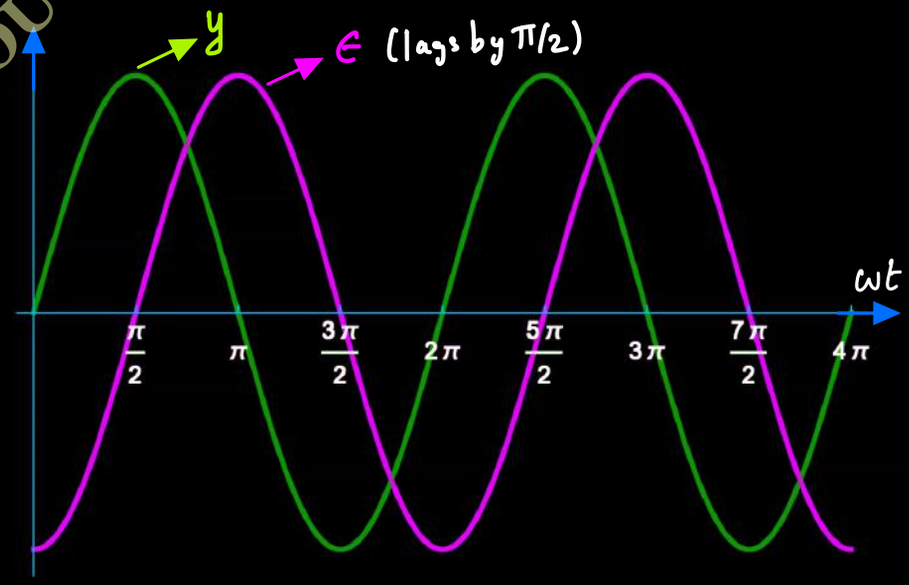
$dV = V'-V = S\,dy$. Thus volumetric strain: $\epsilon = dV/V = dy/dx$.

Thus we have relation between volumetric strain and displacement wave:

$$\epsilon(x) = \frac{dy}{dx} = \frac{dV}{V} \quad (1)$$

Wherever $\epsilon(x)$ is positive, the gas is rarefied and wherever $\epsilon(x)$ is negative, the gas is compressed. Now let $y = A\sin(\omega t - kx)$ then

$\epsilon(x) = -kA\cos(\omega t - kx) = kA\sin(\omega t - kx - \pi/2)$ thus strain wave lags behind the displacement wave by a phase of $\pi/2$ for a right travelling wave. [For a left travelling wave, it would lead by a phase of $\pi/2$).



Displacement And Strain Relation (V +ve)

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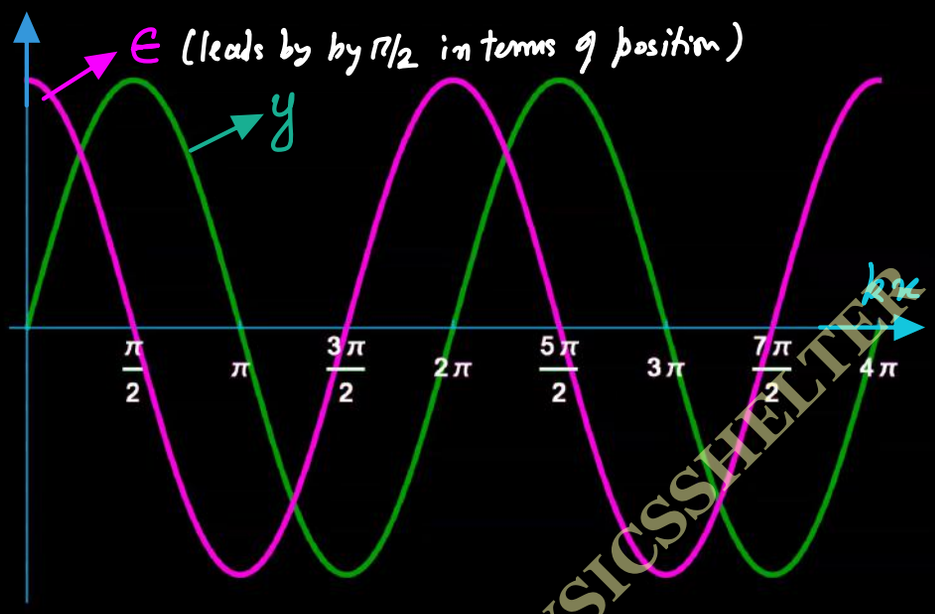
For phase relation with position we consider the displacement wave as $y = A\sin(kx-\omega t) \Rightarrow \epsilon = dy/dx = Ak\cos(kx-\omega t)$

$\Rightarrow \epsilon = Ak\sin(kx-\omega t+\pi/2)$, thus with space, strain wave leads the displacement wave by a phase of $\pi/2$ (Both for right travelling wave as well as left travelling wave)



→ For finding phase relation with time always choose $A\sin(\omega t-kx)$ and for finding relation with space choose $A\sin(kx-\omega t)$.

Relation Between Displacement And Pressure Wave:

Recall from our study of elasticity, $B = -p/(dV/V)$ where p is the excess pressure over and above the ambient pressure (or gauge pressure if surrounding pressure is atmospheric pressure). Thus:

$$p = -B\left(\frac{dV}{V}\right) = -B\frac{\partial y}{\partial x} \quad (2) \text{ (using (1))}.$$

$$\boxed{p = -B\,\partial y/\partial x} \quad (2)$$

Also $B = p/(\rho_e/\rho)$ where ρ_e represents excess density. Thus, $\rho_e = \rho p/B$, thus density wave and pressure wave are always in same phase, (High pressure $\Rightarrow$ High density and low pressure $\Rightarrow$ low density).

Now let $y = A\sin(\omega t-kx) \Rightarrow p = BAk\cos(\omega t-kx)$

$\Rightarrow p = BAk\sin(\omega t-kx+\pi/2)$. Thus for right travelling wave, pressure wave leads the displacement wave by a phase of $\pi/2$.

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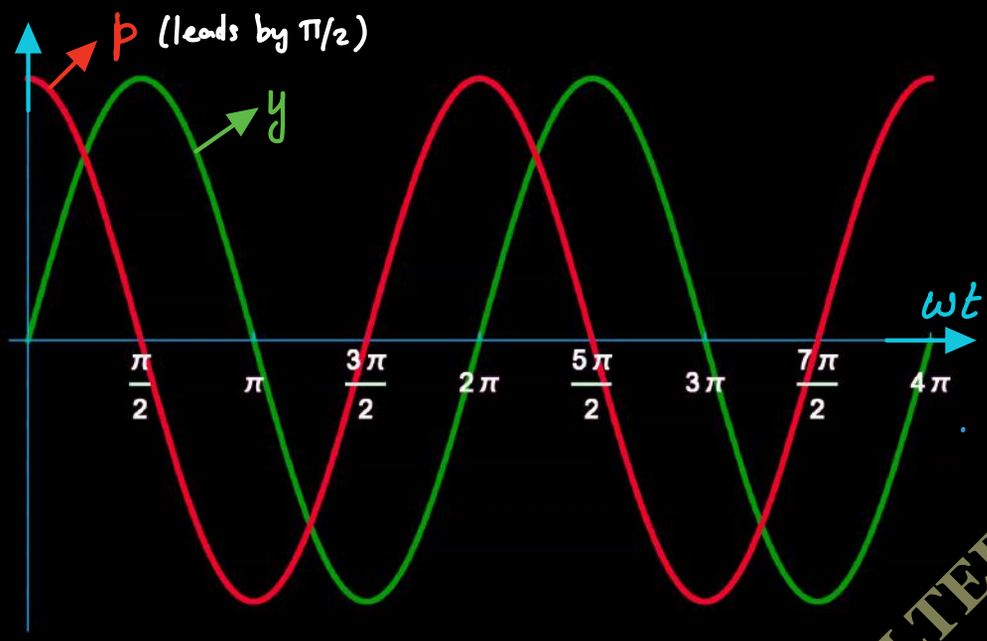
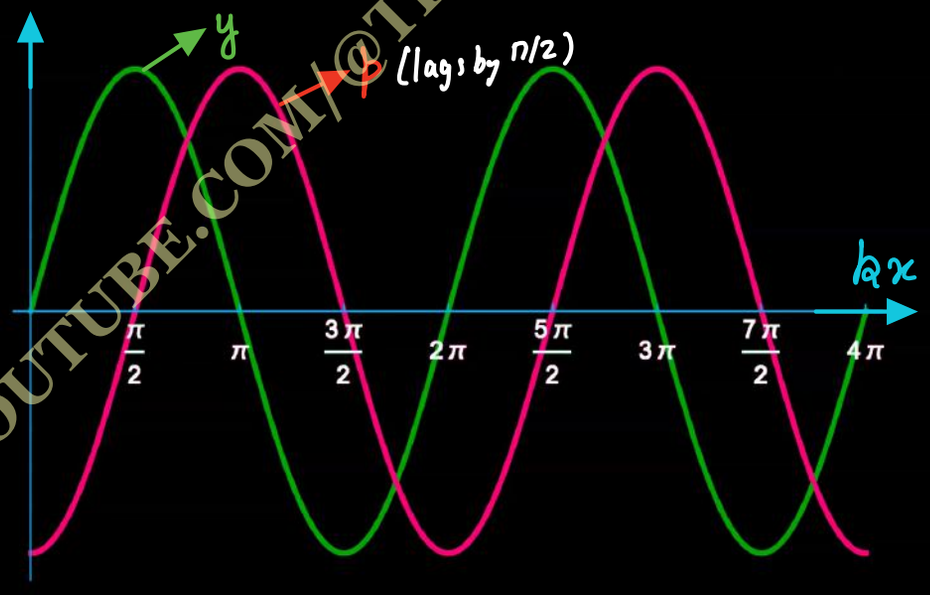
(For left travelling wave, it lags by $\pi/2$).



PRESSURE AND DISPLACEMENT VARIATION WITH TIME (ϑ +Ve)

For finding phase relationship with x coordinate we choose $y = A\sin(kx-\omega t)$.
$\Rightarrow p = -BAk\cos(kx-\omega t) = BAk\sin(kx-\omega t-\pi/2)$. Thus for a right travelling wave, pressure lags by a phase of $\pi/2$ with x coordinate. (This is true for both right and left travelling waves).



DISPLACEMENT AND PRESSURE VARIATIONS WITH x COORDINATE

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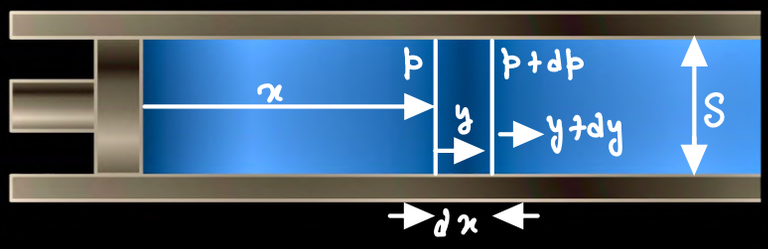
Velocity Of Longitudinal Waves:

Let us apply Newton's second law to the volume element of thickness dx as shown in adjacent figure:



$$pS - (p+dp)S = \rho S\,dx\left(\frac{d^2y}{dt^2}\right)$$

($\because \vec{F} = m\vec{a}$). Rearranging above equation we get:

$$-\frac{dp}{dx} = \rho\frac{d^2y}{dt^2}$$ (However we mean partial derivatives here since p is a multivariable function).

i.e $-\partial p/\partial x = \rho\,\partial^2y/\partial t^2$. Now from ②, $p = -B\,\partial y/\partial x$, thus:

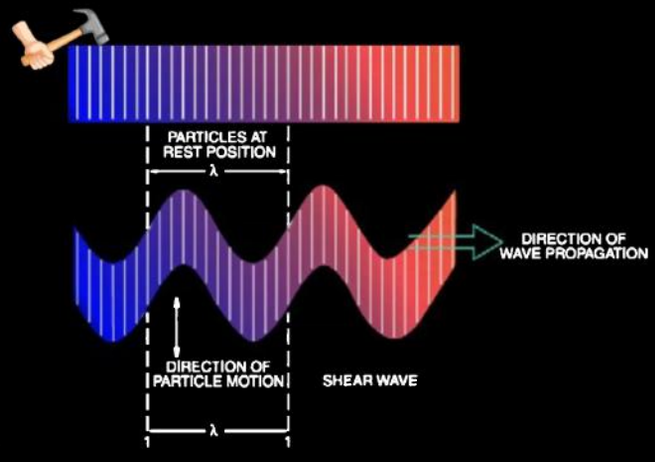
$$-\frac{\partial}{\partial x}\left(-B\frac{\partial y}{\partial x}\right) = \rho\frac{\partial^2 y}{\partial t^2} \Rightarrow \frac{\partial^2 y}{\partial t^2} = \frac{B}{\rho}\frac{\partial^2 y}{\partial x^2} \quad ④$$

Comparing ④ with standard wave PDE: $\frac{\partial^2 y}{\partial t^2} = v^2\frac{\partial^2 y}{\partial x^2}$

$$v = \sqrt{\frac{B}{\rho}}$$ similarly in solids:

$$v = \sqrt{\frac{Y}{\rho}}$$ for longitudinal waves

$$v = \sqrt{\frac{G}{\rho}}$$ for shear waves.



→ For sound waves in gases, B generally means adiabatic Bulk Modulus because compressions and rarefactions occur so fast that, practically no heat transfer takes place during the phenomena thus:

$B = \gamma P = \frac{\gamma PRT}{M}$ thus for gases, $$v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}}$$

→ Here γ is called Laplace correction factor because earlier, Newton had mistakenly assumed that velocity of sound is same as RMS velocity of gas particles (i.e $\sqrt{3RT/M}$), however Laplace corrected this formula.

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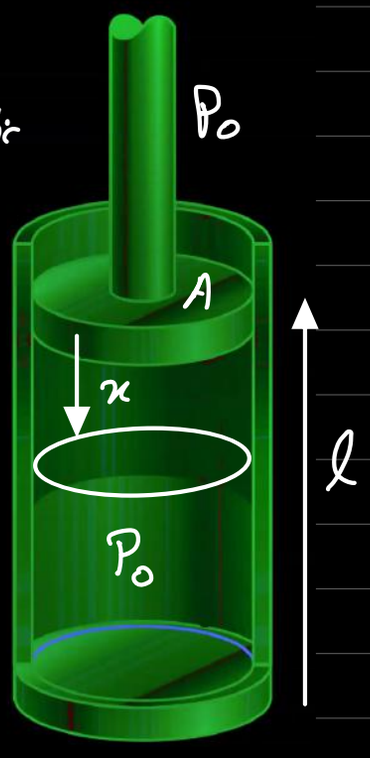
EXAMPLE: In a sinusoidal sound wave maximum pressure variations are 3×10^{-2} Pa above and below atmospheric pressure. Find the displacement amplitude if frequency is 1000 Hz. Given velocity of sound is 344 ms^{-1} and Bulk modulus is 1.42×10^5 Pa.

Solution: $\because$ $p = -B\,\partial y/\partial x$, if $y = A\sin(\omega t - kx)$ $\boxed{p_{max} = BAk}$

$\Rightarrow$ $A = p_{max}/Bk$ also $k = \omega/v$ $\Rightarrow$ $A = p_{max} \times v/B\omega$

$$\Rightarrow A = \frac{p_{max}\, v}{B \times 2\pi f} = \frac{3 \times 10^{-2} \times 344}{1.42 \times 10^5 \times 2\pi \times 1000} = 1.16 \times 10^{-8}\ m = 11.6\ nm \text{ Ans.}$$

AIR SPRING: Imagine air enclosed in a cylinder under a massless piston at atmospheric pressure, with a natural column length l. The cylinder is adiabatic and bulk modulus of air is B. We know that if we push the piston a little, it will try to spring back and we wish to know the equivalent spring constant for the arrangement. Imagine pushing the piston by a distance x and let the pressure of the gas increase by an amount p in the process. x is sufficiently small.



Thus:

$$B = \frac{-p}{(\Delta V/V)} = \frac{-p}{(-Ax)} \times Al$$

$\Rightarrow p = \frac{Bx}{l}$, therefore restoring force on piston: $F = pA = \frac{BAx}{l}$

Now for a spring, $F = kx$, thus by comparison for a fluid spring:

$\boxed{k_{eff} = \frac{BA}{l}}$. The value of B will depend on the nature of process. In case of adiabatic process, $B = \gamma P = \frac{\gamma \rho RT}{M}$ and for isothermal process, $B = P = \frac{\rho RT}{M}$, for polytropic, $B = xP$.

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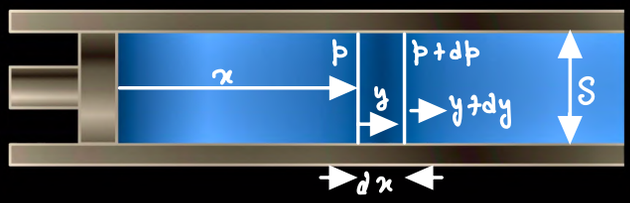
→ This equivalence to a linear spring holds good for small displacements and can be used for time period of SHM of pistons in such arrangements so long as amplitudes are small.

Energy Per Unit Length In Longitudinal Waves:

Consider an element of gas column of length dx being stretched by an amount dy then, potential energy stored in the element is $\frac{1}{2} k_{eff} (dy)^2$ and



$k_{eff} = (BS/dx)$, thus $dU = \frac{1}{2} \cdot \left(\frac{BS}{dx}\right) \cdot (dy)^2$

$$\Rightarrow \frac{dU}{dx} = \frac{1}{2} BS \left(\frac{dy}{dx}\right)^2 = \frac{1}{2} \rho v^2 S \left(\frac{\partial y}{\partial x}\right)^2 = \frac{1}{2} \mu v^2 \left(\frac{\partial y}{\partial x}\right)^2 \quad [\because \rho S = \mu]$$

Now kinetic energy of the element $dKE = \frac{1}{2} \mu dx (\partial y/\partial t)^2$ where $\mu = \rho S$. Thus $\frac{dKE}{dx} = \frac{1}{2} \mu \left(\frac{\partial y}{\partial t}\right)^2$ ②

From ① & ②, total energy per unit length:

$$e_l = \frac{dE}{dx} = \frac{dKE}{dx} + \frac{dU}{dx} = \frac{1}{2} \mu \left[\left(\frac{\partial y}{\partial t}\right)^2 + v^2 \left(\frac{\partial y}{\partial x}\right)^2 \right]$$

$$\Rightarrow \boxed{e_l = \frac{1}{2} \mu G}, \text{ Here } \boxed{G = \left(\frac{\partial y}{\partial t}\right)^2 + v^2 \left(\frac{\partial y}{\partial x}\right)^2} \qquad \boxed{u = \frac{e_l}{S} = \frac{1}{2} \rho G}$$

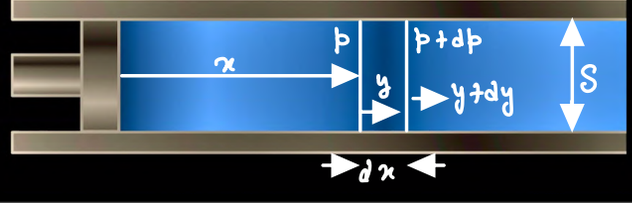
Here u is energy per unit volume.

We see that formula for energy per unit length, Kinetic energy/unit length and potential energy per unit length are exactly same as those obtained for transverse waves. This similarity also holds for Power, Intensity and Energy per unit volume.

Power Developed By Excess Pressure:

$$P = \vec{F} \cdot \vec{v} = pS \frac{\partial y}{\partial t} = -B \frac{\partial y}{\partial x} \cdot S \left(\frac{\partial y}{\partial t}\right)$$

Thus $\boxed{P = -BS \left(\frac{\partial y}{\partial x}\right)\left(\frac{\partial y}{\partial t}\right)}$



This immediately reminds us of power formula in transverse

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waves i.e $P = -T \frac{\partial y}{\partial x} \cdot \frac{\partial y}{\partial t}$. Thus here BS is the analogue of string tension.

→ The formula derived above hold for all kinds of one dimensional longitudinal waves (progressive, stationary or mixed).

INTENSITY OF LONGITUDINAL WAVES: Recall that Intensity is defined as power transmitted per unit cross sectional area. Thus,

$$I = \frac{P}{S} = -B \frac{\partial y}{\partial x} \cdot \frac{\partial y}{\partial t} \qquad \boxed{I = -B \frac{\partial y}{\partial x} \frac{\partial y}{\partial t}}$$

ENERGY DENSITY, POWER AND INTENSITY FOR SINUSOIDAL PROGRESSIVE WAVES:

Just as for transverse waves, all formulae for longitudinal progressive waves are analogous to those of transverse waves.

→ Just as for transverse waves, kinetic energy per unit length is equal to potential energy per unit length for progressive longitudinal waves. We state all the results here without formal derivations (which are exactly as done for transverse waves).

Thus for progressive sinusoidal waves: (Assumed $y = A\sin(kx - \omega t)$)

$$\boxed{\rho_l = \mu A^2 \omega^2 \cos^2(kx - \omega t)} \qquad \boxed{U_l = \rho A^2 \omega^2 \cos^2(kx - \omega t)}$$

$$\boxed{\langle \rho_l \rangle = \frac{1}{2} \mu A^2 \omega^2} \qquad \boxed{\langle U \rangle = \frac{1}{2} \rho A^2 \omega^2}$$

$$\boxed{P = \mu v A^2 \omega^2 \cos^2(kx - \omega t)} \qquad \boxed{I = \rho v A^2 \omega^2 \cos^2(kx - \omega t)}$$

$$\boxed{\langle P \rangle = \frac{1}{2} \mu v A^2 \omega^2} \qquad \boxed{\langle I \rangle = \frac{1}{2} \rho v A^2 \omega^2}$$

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Intensity In Terms Of Excess Pressure Amplitude:

We have $I_{avg} = \frac{1}{2}\rho v A^2\omega^2$ and also pressure amplitude $p_m = BAk$

also $v = \sqrt{B/\rho}$ or $B = \rho v^2$, thus $p_m = \rho v^2 A(\omega/v) = \rho v A\omega$

Or $A\omega = p_m/\rho v$. Using this in the formula for Intensity:

$$I_{avg} = \frac{1}{2}\rho v\left(\frac{p_m}{\rho v}\right)^2 = \frac{p_m^2}{2\rho v}$$

Thus $\boxed{I_{avg} = p_m^2/2\rho v}$ Here p_m is excess pressure amplitude.

Example: A certain single tone sound wave has pressure amplitude 3×10^{-2} Pa and a frequency 1000 Hz. The Bulk modulus is 1.42×10^5 Pa and speed of sound is 344 ms^{-1}. What would be the pressure and displacement amplitudes of another single tone sound of 20 Hz to have same intensity as the 1000 Hz sound?

Solution: Recall that $I = p_m^2/2\rho v$, now for both waves ρ and v are same ($v = \sqrt{B/\rho} = \sqrt{\gamma P_0/\rho}$). Thus p_m must be same for both the waves to have same intensity. Thus $p_m = 3\times10^{-2}$ Pa Ans.

Now $p_m = BAk \Rightarrow A = p_m/Bk$ also $k = \omega/v$

$$\Rightarrow A = \frac{p_m \times v}{B\omega} = \frac{p_m \times v}{2\pi f B} = \frac{3\times10^{-2}\times 344}{2\pi\times 20\times 1.42\times10^5} = 5.78\times10^{-7}\text{ m}$$

thus $A \approx 0.6\,\mu$m Ans.

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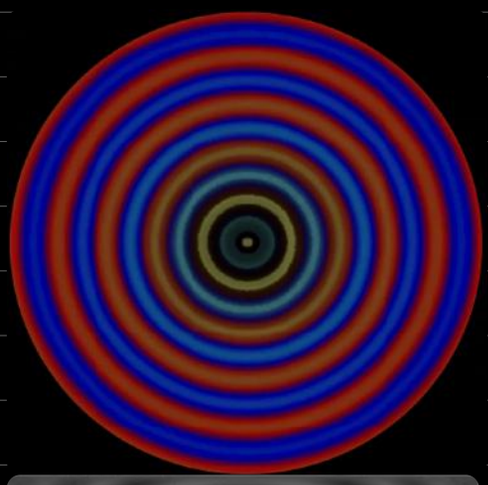
Spherically Symmetric Isotropic Sound Propagation:

A spherical isotropic emitter emits sound equally in all directions. If P is the power of the source then by the time it reaches a radius r, the power gets distributed over an area $4\pi r^2$. Thus intensity of such a source declines according to:

$$I = \frac{P}{4\pi r^2} \quad \text{(inverse square declining)}.$$



Now since intensity is proportional to square of amplitude ($I = \frac{1}{2}\rho v A^2\omega^2$) thus as the sound propagates, the amplitude declines with $1/r$. A typical isotropic spherical wave can be represented as:

$$\Psi(r,t) = A\frac{r_0}{r}\sin(kr - \omega t)$$

Here A is the amplitude at a distance r_0 from the centre of the sphere.

Wavefront: A surface joining particles oscillating in same phase is called a wavefront. For example the picture and animation above shows the wavefront propagation with time. The direction of propagation of disturbance is always perpendicular to the wavefronts. For spherical isotropic source, wavefronts are concentric spheres (moving outwards with time). Thus wave velocity is also known as phase velocity.

Example: Irodov 4.158:

Find the radius vector defining the position of a point source of spherical waves if that source is known to be located on the straight line between the points with radius vectors $\vec{r}_1$ and $\vec{r}_2$ at which the oscillation amplitudes of particles of the medium are equal to a_1 and a_2. The damping of the wave is negligible, the medium is homogeneous.

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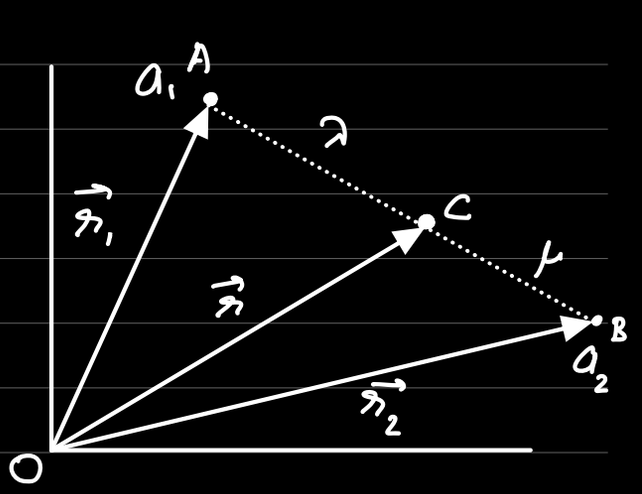
Solution: Amplitude declines according to inverse ratio of distances from the source.

Thus $CA \propto \frac{1}{a_1}$ and $CB \propto \frac{1}{a_2}$

Thus $\frac{CA}{CB} = \frac{\lambda}{\mu} = \frac{a_2}{a_1}$. Thus using section formula, $\vec{r} = \frac{a_2\vec{r}_2 + a_1\vec{r}_1}{a_1 + a_2}$ Ans.



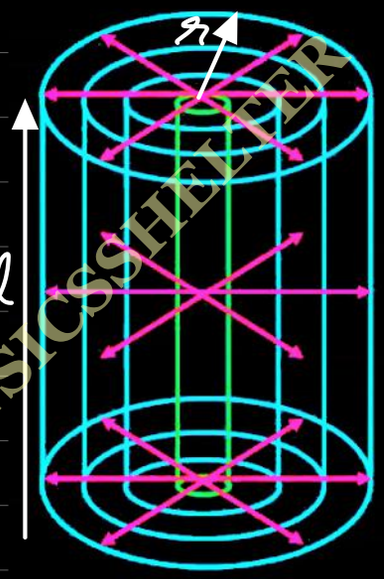
CYLINDRICALLY SYMMETRIC PROPAGATION:

Imagine a line shaped pulsating woofer emanating sound in all directions with cylindrical symmetry. If P is the power of the woofer, by the time the wave spreads to a radius r, the power gets distributed over a surface area $2\pi r l$. Thus intensity is given as:

$$I = P/S = P/2\pi r l$$

$\boxed{I = P/2\pi r l}$. Thus the intensity declines according to $1/r$ and since intensity is proportional to square to amplitude thus, amplitude declines as $1/\sqrt{r}$. A typical cylindrical propagation can be represented as:

$$\psi(r,t) = A\sqrt{\frac{r_0}{r}}\sin(kr - \omega t)$$

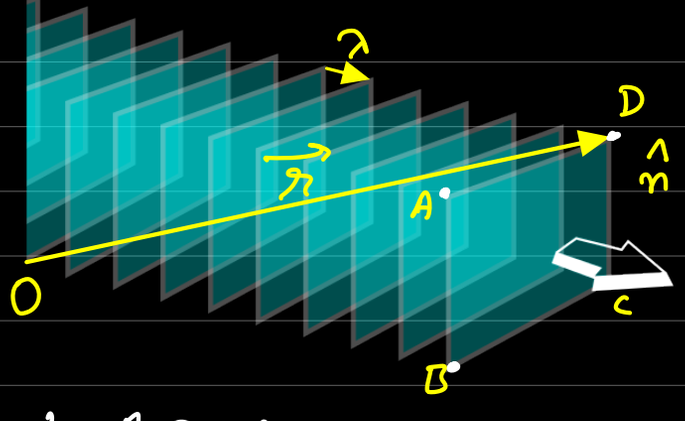


Cylindrical Wavefront from a linear source

here A represents the amplitude at $r = r_0$.

PLANE WAVE ALONG ARBITRARY DIRECTION COSINE:

A wave in which wavefronts are planar is called a planar wave. Consider a plane wave travelling along some direction $\hat{n}$ (not necessarily x, y or z axes, we are referring the direction of propagation as $\hat{n}$).



Now one can readily see that points A, B, C, D

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are all oscillating in same phase. Thus phase does not depend on exact value of $\vec{r}$ rather it depends on its projection along $\hat{n}$. or $\vec{r}\cdot\hat{n}$. When the wave travels along x axis, we would typically write the wave as $A\sin(kx-\omega t)$ etc. Here we must accordingly replace x by $\vec{r}\cdot\hat{n}$. Thus our wave for an instance might look like:

$$\psi = A\sin(k\,\vec{r}\cdot\hat{n} - \omega t) \text{ or } A\sin(\vec{k}\cdot\vec{r} - \omega t)$$

where $\vec{k} = k\hat{n}$ is called wave propagation vector, or just wave vector. Thus a wave travelling along arbitrary direction cosine will take the form:

$$\boxed{\psi = A\sin(\vec{k}\cdot\vec{r} \pm \omega t \pm \phi)} \text{ or}$$

$$\psi = A\cos(\vec{k}\cdot\vec{r} \pm \omega t \pm \phi)$$

EXAMPLE: IRODOV 4.151:

A plane harmonic wave with frequency ω propagates at a velocity v in a direction forming angles α, β, γ with the x, y, z axes. Find the phase difference between the oscillations at the points of medium with coordinates x_1, y_1, z_1 and x_2, y_2, z_2.

Solution: Here $\hat{n} = \cos\alpha\,\hat{i} + \cos\beta\,\hat{j} + \cos\gamma\,\hat{k}$. Let the required wave be:

$\psi = A\sin(\vec{k}\cdot\vec{r} - \omega t)$, phase difference between two positions at a given time is given by $\Delta\phi = \vec{k}\cdot(\vec{r}_2 - \vec{r}_1)$

$$= \frac{\omega}{v}\,\hat{n}\cdot(\vec{r}_2 - \vec{r}_1)$$

$$= \frac{\omega}{v}(\cos\alpha\,\hat{i} + \cos\beta\,\hat{j} + \cos\gamma\,\hat{k})\cdot[(x_2-x_1)\hat{i} + (y_2-y_1)\hat{j} + (z_2-z_1)\hat{k}]$$

$$\Rightarrow \Delta\phi = \frac{\omega}{v}[(x_2-x_1)\cos\alpha + (y_2-y_1)\cos\beta + (z_2-z_1)\cos\gamma] \text{ ans.}$$

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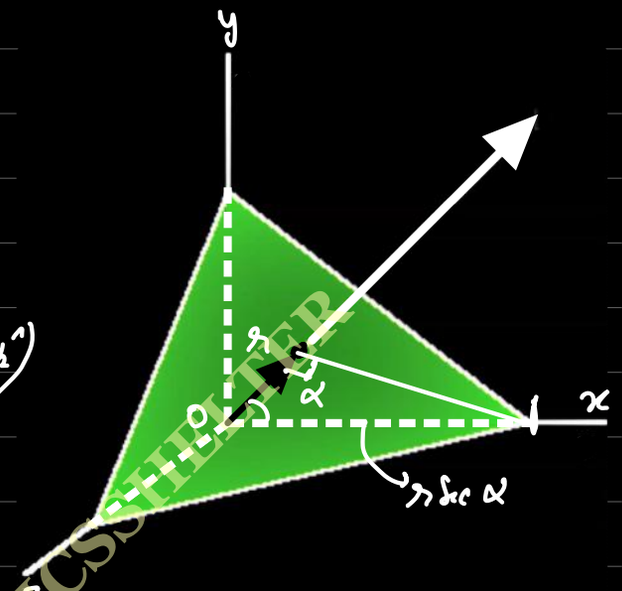
Example: Irodov 4.152:

A plane wave of frequency ω propagates so that a certain phase of oscillation moves along the x, y, z axes with velocities v_1, v_2, v_3 respectively. Find the wave vector $\mathbf{k}$, assuming the unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$ of the coordinate axes to be assigned.

Solution: Let a particular wave front plane be located at a distance r from the origin, then its x, y and z intercepts are $x = r\sec\alpha$, $y = r\sec\beta$ and $r\sec\gamma$.



Now wave vector $= \frac{\omega}{v}(\cos\alpha\,\hat{i} + \cos\beta\,\hat{j} + \cos\gamma\,\hat{k})$

Now $\because x = r\sec\alpha \Rightarrow v_1 = v\sec\alpha$

$\Rightarrow \cos\alpha = \frac{v}{v_1}$ similarly $\cos\beta = \frac{v}{v_2}$ and

$\cos\gamma = \frac{v}{v_3}$. Thus $\vec{k} = \frac{\omega}{v}\left[\frac{v}{v_1}\hat{i} + \frac{v}{v_2}\hat{j} + \frac{v}{v_3}\hat{k}\right]$

$\Rightarrow \vec{k} = \omega\left[\frac{1}{v_1}\hat{i} + \frac{1}{v_2}\hat{j} + \frac{1}{v_3}\hat{k}\right]$

Damped Wave In A Homogeneously Absorbing Medium: A homogeneously absorbing medium is defined in which fractional energy absorbed per unit length of propagation remains constant. For such media, the amplitude is smaller than the case of a non dissipatin medium. For such media we can write the wave equations as follows:

$$\boxed{y = Ae^{-\gamma(x-x_0)}\sin(kx-\omega t)}$$ for plane waves and

$$\boxed{y = A\frac{r_0}{r}e^{-\gamma(r-r_0)}\sin(kr-\omega t).}$$

Here γ is called wave damping coefficient.

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Example: Irodov 4.159

A point isotropic source generates sound oscillations with frequency $\nu = 1.45$ kHz. At a distance $r_0 = 5.0$ m from the source the displacement amplitude of particles of the medium is equal to $a_0 = 50\ \mu$m, and at the point A located at a distance $r = 10.0$ m from the source the displacement amplitude is $\eta = 3.0$ times less than a_0. Find :

(a) the damping coefficient γ of the wave;

(b) the velocity oscillation amplitude of particles of the medium at the point A.

Solution: A damped spherical wave has form $\psi = a_0 \frac{r_0}{r} e^{-\gamma(r-r_0)} \sin(kr-\omega t)$

Now according to our problem, $\frac{a_0 r_0}{r} e^{-\gamma(r-r_0)} = \frac{1}{\eta} a_0$

$\Rightarrow e^{-\gamma(r-r_0)} = r/r_0\eta \Rightarrow -\gamma(r-r_0) = \ln(r/r_0\eta)$ or

$$\gamma = \frac{1}{r-r_0} \ln(\eta r_0/r) \text{ Ans.}$$

Now velocity amplitude = Displacement amplitude $\times \omega$

$$\Rightarrow v_{max} = \frac{a_0}{\eta}\omega = \frac{2\pi\nu a_0}{\eta} \text{ Ans.}$$

Example: Irodov 4.160:

Two plane waves propagate in a homogeneous elastic medium, one along the x axis and the other along the y axis; $\xi_1 = a\cos(\omega t - kx)$, $\xi_2 = a\cos(\omega t - ky)$. Find the wave motion pattern of particles in the plane xy if both waves

(a) are transverse and their oscillation directions coincide;

(b) are longitudinal.

Solution: (a) Direction of oscillation is $\perp$ to the direction of propagation thus, here it must be simultaneous $\perp$ to x & y axes i.e it must be along $\hat{k}$. Thus $\psi = \xi_1 + \xi_2 = a\hat{k}[\cos(\omega t - kx) + \cos(\omega t - ky)]$

$$\Rightarrow \vec{\psi} = 2\cos\frac{k}{2}[x-y]\cos\left[k\left(\frac{x+y}{2}\right) - \omega t\right]\hat{k}$$

$$\Rightarrow \vec{\psi} = 2\cos\frac{k}{2}(x-y)\cos\left(k\left(\frac{\hat{i}+\hat{j}}{2}\right)\cdot\vec{r} - \omega t\right)\hat{k}$$

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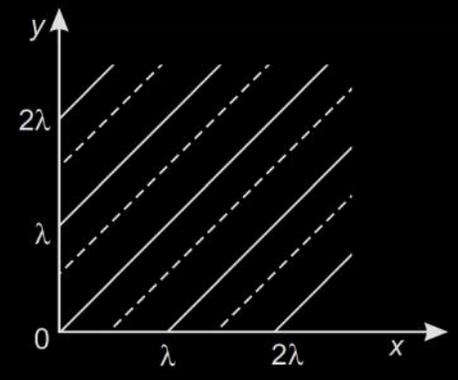
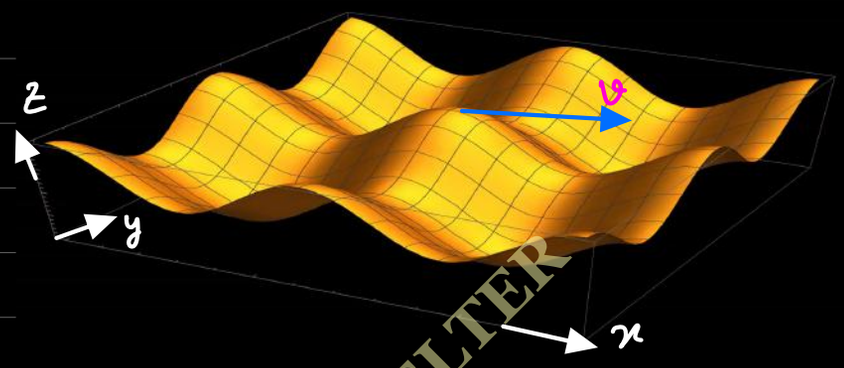
Here the first term $2\cos\frac{k}{2}(x-y)$ tells us that amplitude of the oscillation depends on (x,y) coordinate of the particle. Thus particles along the line $x-y=$ constant all have same amplitudes of oscillations and the direction of propagation is along the unit vector $(\hat{i}+\hat{j})/\sqrt{2}$.





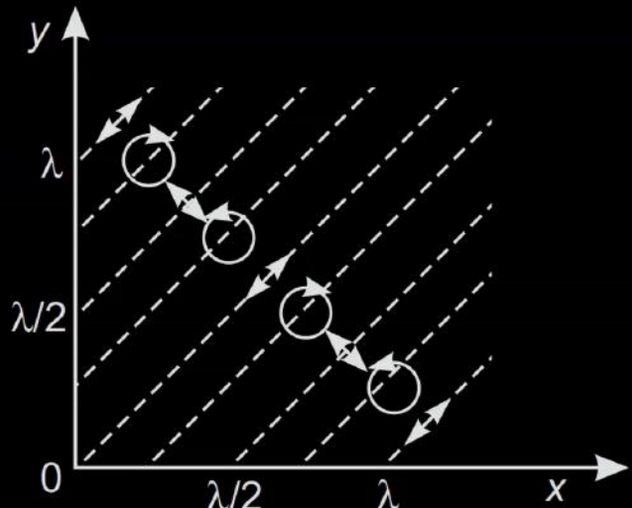
(b) If both waves are longitudinal we have,

$$\vec{\xi}_1 = a\cos(\omega t - kx)\,\hat{i} \quad \text{and} \quad \vec{\xi}_2 = a\cos(\omega t - ky)\,\hat{j}$$

Thus motion of any particle at (x,y) is given as:

$$\vec{\psi} = \vec{\xi}_1 + \vec{\xi}_2 = a\left[\cos(\omega t - kx)\,\hat{i} + \cos(\omega t - ky)\,\hat{j}\right]$$

Now recall that superposition of two mutually perpendicular motions leads to an elliptical motion except when the phase difference is 0 or π. Thus particles when $(\omega t - ky) - (\omega t - kx) = n\pi$ such particles will move along straight lines i.e particles oscillating in straight lines are located at $y - x = n\pi/k$. For even n, the particles oscillate along $y = x$ line and for odd n particles oscillate along $y = -x$ direction. For $y - x = (2n+1)\frac{\pi}{2}$, the particles rotate in a circle. For even n, we get anticlockwise circle and for odd n we will get clockwise circles as per theory of superposition mutually perpendicular SHMs.



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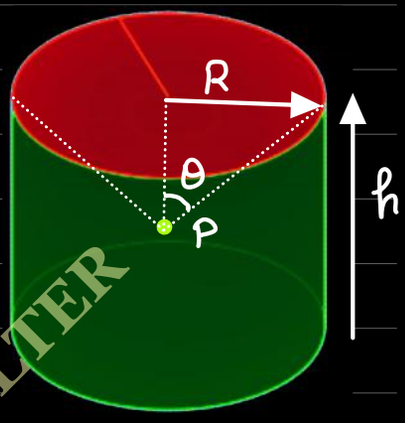
Example : Irodov 4.162 :

A point isotropic source with sonic power $P = 0.10$ W is located at the centre of a round hollow cylinder with radius $R = 1.0$ m and height $h = 2.0$ m. Assuming the sound to be completely absorbed by the walls of the cylinder, find the mean energy flow reaching the lateral surface of the cylinder.



Solution: Power per unit solid angle : $P/4\pi$

Power from flat lids : $\frac{P}{4\pi} \times 2\pi(1-\cos\theta) \times 2 = P(1-\cos\theta)$

$\left[\because\ \omega_{lid} = 2\pi(1-\cos\theta)\right]$, Now $P_{CSA} = P - P_{lids} = P\cos\theta$

$$\Rightarrow P_{CSA} = \frac{P\,h/2}{\sqrt{R^2 + (h/2)^2}} = \frac{Ph}{\sqrt{h^2 + 4R^2}} \text{ Ans.}$$

Loudness: Loudness is a measure of Intensity of a sound wave on a logarithmic scale. We define loudness as:

$$\beta = \log_{10}\left(\frac{I}{I_0}\right) \text{ Bels or}$$

$$\boxed{\beta = 10\log_{10}\left(\frac{I}{I_0}\right)} \text{ decibels (dB)}$$

Here I_0 is reference intensity of 10^{-12} W/m^2. This is the minimum intensity audible to human beings at a frequency of 1 KHz.

Example : Irodov 4.188:

At a distance $r = 100$ m from a point isotropic source of sound of frequency 200 Hz the loudness level is equal to $L = 50$ dB. The audibility threshold at this frequency corresponds to the sound intensity $I_0 = 0.10$ nW/m^2. The damping coefficient of the sound wave is $\gamma = 5.0 \times 10^{-4}$ m^{-1}. Find the sonic power of the source.

Solution: $L = 10\log\left(\frac{I}{I_0}\right) \Rightarrow I = I_0 \times 10^{L/10} = \frac{P}{4\pi r^2} e^{-2\gamma r}$

$$\Rightarrow P = 4\pi r^2 I_0 e^{2\gamma r} \times 10^{L/10} = 1.4\text{ W} \quad \text{(if } L \text{ is measured in decibels)}$$

$$\text{or } P = 4\pi r^2 I_0 e^{2\gamma r} \times 10^{L} \quad \text{(if } L \text{ is measured in Bels).}$$

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Note: For JEE purposes always use $I_0 = 10^{-12}\ W/m^2$ (Even though here Irodov has specified a different value of I_0).

Example: Irodov 4.182:

A plane sound wave propagates along the x-axis. The damping coefficient of the wave is $\gamma = 0.0230\ m^{-1}$. At the point $x = 0$ the loudness level is $L = 60$ dB. Find :

(a) the loudness level at a point with coordinate $x = 50$ m;

(b) the coordinate x of the point at which the sound is not heard any more.

Solution: $I = I_{x_0} e^{-2\gamma(x-x_0)}$ (∵ amplitude declines as $e^{-\gamma(x-x_0)}$)

At $x = 0$, $L = 10 \log_{10}(I_{x_0}/I_0)$

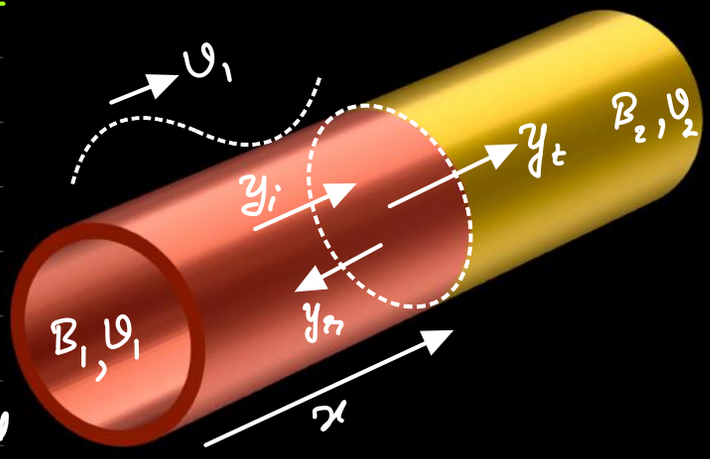
at $x = 50$, $L' = 10 \log_{10}(I_{x_0} e^{2\gamma x}/I_0) = L - 20\gamma x \log_{10} e$

$\Rightarrow L' = L - 20\gamma x \log_{10} e = 50$ dB (in decibels).

Now for sound to become inaudible, $L' = 0 \Rightarrow x = L/20\gamma \log_{10} e = 0.3$ Km.

Reflection & Transmission Of Sound Waves:

Let two gases ① & ② be separated by a massless partition. Further let Gas ① have B_1 and v_1 as Respectin Bulk modulus and velocity and B_2 and v_2 are the corresponding quantities for gas 2. To find the reflected and transmitted disturbances we shall use the following conditions:



Ⓘ Continuity of displacement i.e at the boundary $y_i + y_r = y_t$ where y_i, y_r and y_t are incident, reflected and transmitted disturbances respectively.

Ⓘⓘ Since partition is massless, pressure to the left and right of it must be equal i.e $p_i + p_r = p_t$ at the boundary.

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$y_i = f_i(x - v_1 t)$, $y_r = f_r(x + v_1 t)$, $y_t = f_t(x - v_2 t)$.

Now from condition (I), $y_i + y_r = y_t$ (1) (At the boundary)

From condition (II) $-B_1 \frac{\partial y_i}{\partial x} - B_1 \frac{\partial y_r}{\partial x} = -B_2 \frac{\partial y_t}{\partial x}$ (2) ($\because p = -B \partial y/\partial x$)

$\Rightarrow \frac{-B_1}{v_1} \dot{y}_i + \frac{B_1}{v_1} \dot{y}_r = -\frac{B_2}{v_2} \dot{y}_t$ (3) ($\because \frac{\partial y}{\partial t} = -v \frac{\partial y}{\partial x}$)

From (1):

$$\dot{y}_i + \dot{y}_r = \dot{y}_t \quad (4)$$

Now in equations (3) & (4) we can treat $\dot{y}_r$ & $\dot{y}_t$ as unknowns and solve for them in terms of $\dot{y}_i$. We get:

$$\dot{y}_r = \left(\frac{v_2/B_2 - v_1/B_1}{v_2/B_2 + v_1/B_1}\right) \dot{y}_i \text{ and } \dot{y}_t = \frac{2 v_2/B_2}{v_1/B_1 + v_2/B_2} \cdot \dot{y}_i$$

Integrating with time and using the fact that when $y_i = 0$, $y_r = y_t = 0$. Thus:

$$\boxed{y_r = \left(\frac{v_2/B_2 - v_1/B_1}{v_2/B_2 + v_1/B_1}\right) y_i} \quad (A) \qquad \text{and} \qquad \boxed{y_t = \left(\frac{2 v_2/B_2}{v_2/B_2 + v_1/B_1}\right) y_i} \quad (B)$$

If $B_1 = B_2$ (for instance both sides is diatomic gas) then these formulae reduce to the formulae that we derived for transverse waves.

Reflected And Transmitted Pressure Waves:

At the boundary, $p_i + p_r = p_t$ (5) Also from (II):

$-v_1 y_i' + v_1 y_r' = v_2 y_t'$, Now $p = -By' \Rightarrow y' = -\frac{1}{B} p$

$$\Rightarrow \frac{v_1}{B_1} p_i - \frac{v_1}{B_1} p_r = \frac{v_2}{B_2} p_t \quad (6)$$

$$\Rightarrow \frac{1}{\rho_1 v_1} p_i - \frac{1}{\rho_1 v_1} p_r = \frac{1}{\rho_2 v_2} p_t \quad (7)$$

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Solving (5) and (7) for p_r and p_t:

$$p_r = \frac{\rho_2 v_2 - \rho_1 v_1}{\rho_2 v_2 + \rho_1 v_1} p_i \quad \text{(C)} \qquad \text{and} \qquad p_t = \frac{2\rho_2 v_2}{\rho_1 v_1 + \rho_2 v_2} p_i \quad \text{(D)}$$

→ Consider equations (A) and (C), if at any boundary displacement wave undergoes a π shift then the pressure wave will not undergo a π shift and if displacement wave does not undergo a π shift then pressure wave will undergo a π shift. Here's the proof:

From equation (A) For π shift in displacement wave:

$$v_2/B_2 - v_1/B_1 < 0 \implies B_2/v_2 - B_1/v_1 > 0$$

($\because$ v, B are all positive).

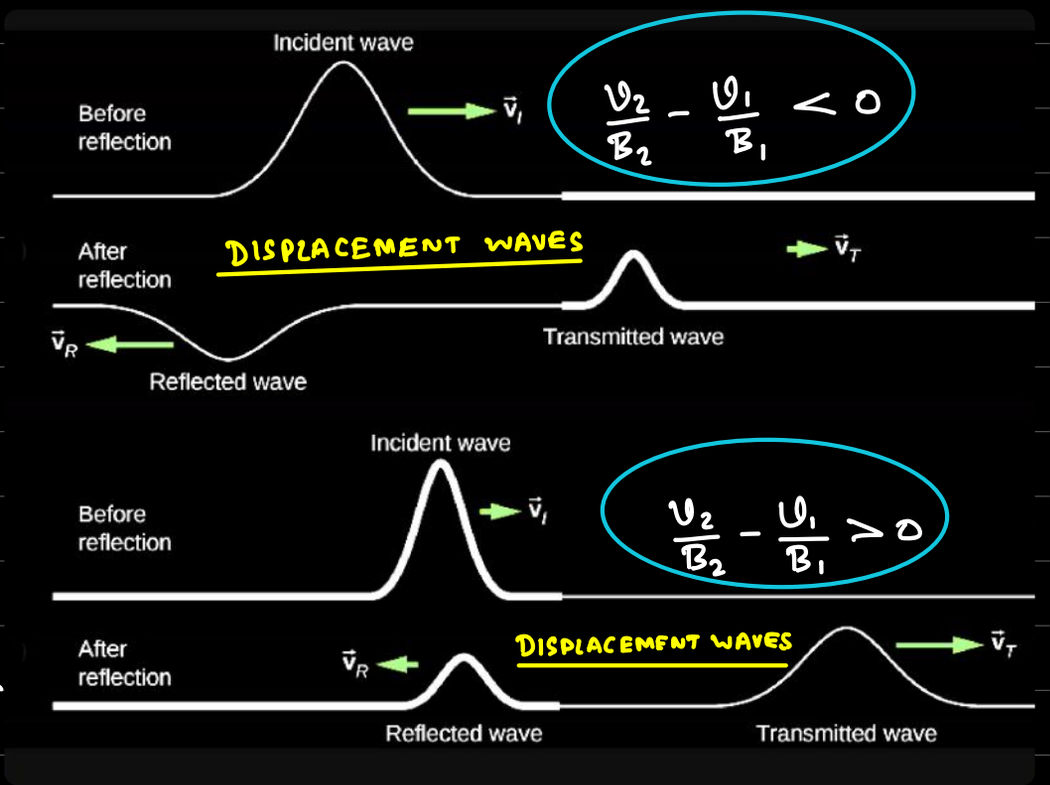
$\implies \rho_2 v_2^2/v_2 - \rho_1 v_1^2/v_1 > 0 \implies \rho_2 v_2 - \rho_1 v_1 > 0$ thus pressure wave will not undergo any π shift. (See equation (C)).

→ Whether or not a sound wave will undergo a π shift depends on the difference $v_2/B_2 - v_1/B_1$ rather than just $v_2 - v_1$

(For instance at water air interface, $v_{water} > v_{air}$ but still when wave from air strikes water interface, the displacement wave undergoes a π shift, because $v_w/B_w - v_a/B_a < 0$).

The adjacent diagram visually depicts how displacement waves will transmit across a boundary.

For pressure waves the diagrams will be swapped. For solids similar diagrams can be obtained for stress waves. Substitute $v/B \to v/Y$.



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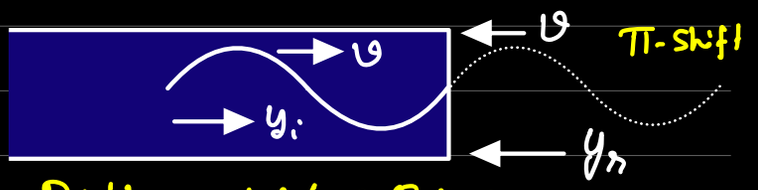
Reflection From Closed End Of A Pipe:

If a wave is incident towards closed end of a pipe then at the closed end net displacement must be zero, thus $y_i + y_r = 0 \Rightarrow y_r = -y_i$. or we can say that incident wave undergoes a π shift and returns back with same amplitude. Entire energy is reflected back and nothing is transmitted. For modelling reflection of displacement waves from closed end, we take the mirror image of water image in the closed end for the reflected wave and superpose to get net displacement.



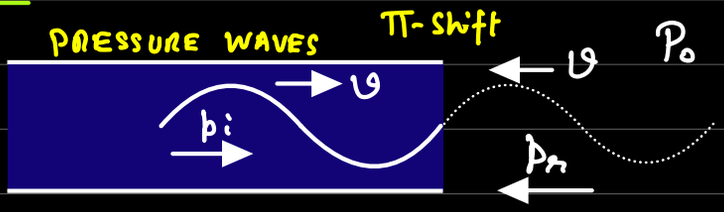
Displacement Wave Reflection.

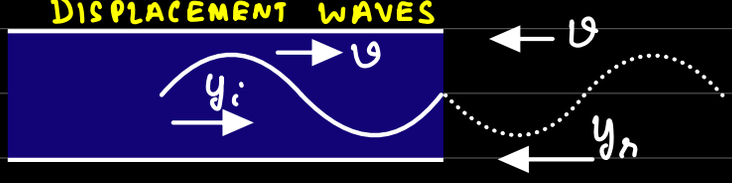
Since displacement wave undergoes a π shift, the pressure wave does not undergo any π shift. (at the closed end, the boundary behaves as if $B \to \infty$). Thus for modelling of reflection of pressure waves, we simply take the mirror image of pressure wave and net wave is obtained by superposing the reflected and incident waves.

Pressure Wave Reflection

Reflection From Open End Of A Pipe:

The open end is open to the atmosphere thus pressure here is always atmospheric and hence excess pressure is zero. (Boundary open to atmosphere can be considered as the boundary across which gas undergoes isobaric process thus PV^0 = Constant and therefore bulk modulus is zero (Recall polytropic Bulk modulus = γP). Thus at the open end, $p_i + p_r = 0$ or $p_r = -p_i$. thus the pressure wave undergoes a π shift at the open end and therefore the displacement wave undergoes no π shift. For geometric modelling of reflected wave, for displacement wave we simply take mirror image and for pressure wave we take water image of mirror image.





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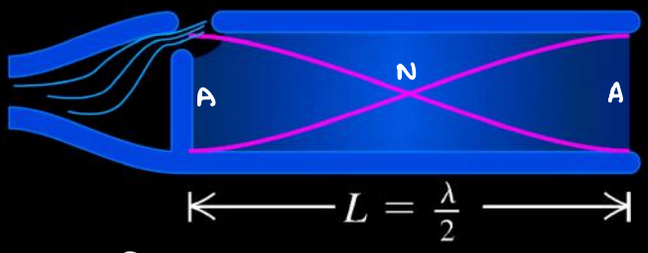
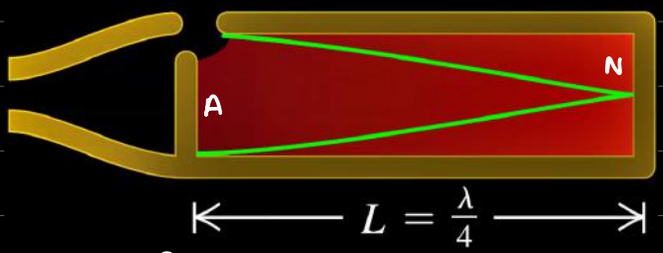
STATIONARY WAVES IN ORGAN PIPES / WIND INSTRUMENTS :



PIPE OPEN AT BOTH ENDS



PIPE CLOSED AT ONE END

CHURCH ORGAN

Several wind instruments such as organs, flutes, trumpets produce musical sounds using stationary waves in air columns. The stationary waves are produced by superposition of incident waves and reflected waves from either open end or a closed end. The air column is typically excited in to vibrations by blowing air in a mouthpiece (which might sometimes contain vibrating reed of wood or some other material like in case of a shehnai, Nadaswaram or Saxophone, in case of trumpets, lips themselves substitute for the reed and the air has to be blown in a certain way to get the desired vibration). As described in previous section, the open end forms a displacement antinode (or a pressure node) and the closed end forms the displacement node (or pressure antinode). Different notes are produced by varying the effective length of the pipe either by closing and opening of valves or opening or closing of holes.

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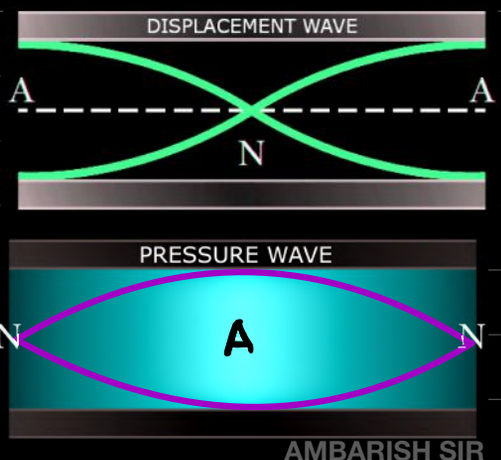
STATIONARY WAVES IN PIPE OPEN AT BOTH ENDS:

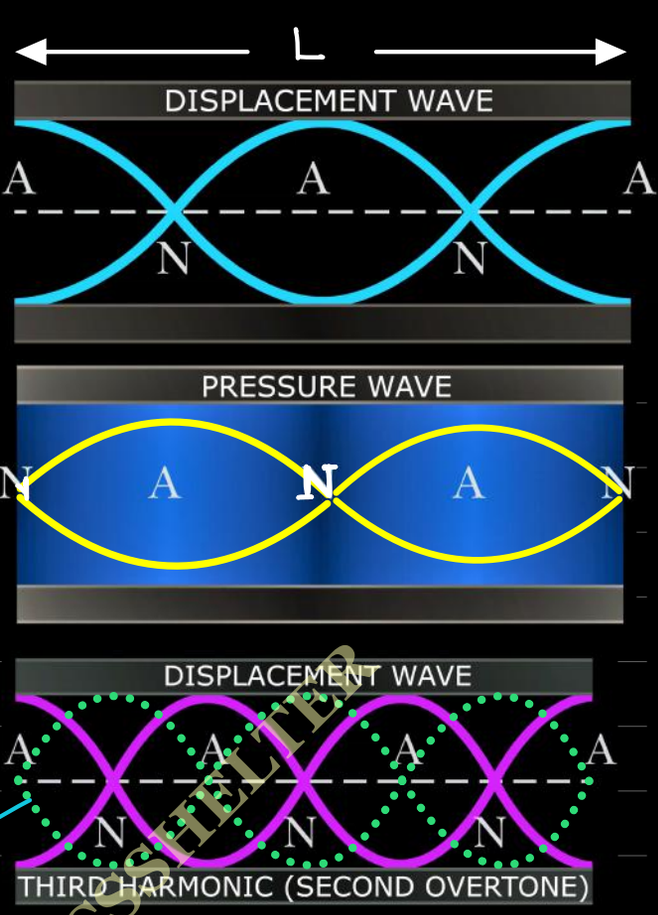
FUNDAMENTAL MODE: This is the minimum frequency for forming well defined stationary waves of large amplitudes. Recall that in stationary waves, antinode to antinode distance is $\lambda/2$. Thus here if L is length of the pipe, $L = \lambda/2 = v/2\nu \Rightarrow \boxed{\nu = v/2L}$



FIRST OVERTONE (SECOND HARMONIC):

Here as clear from adjacent figure,
$L = 2\lambda/2 = 2v/2\nu \Rightarrow \nu = 2v/2L$.
This pattern can be continued further for higher harmonics also thus we can write for third harmonic (or second overtone), $\nu = 3v/2L$ and similarly for n^{th} harmonic ($n-1^{st}$ overtone as):

$$\boxed{\nu = nv/2L}$$



HARMONIC NUMBER:

Any resonance frequency divided by fundamental frequency of the system is called harmonic number. For example if 50 Hz is fundamental frequency of a system then 250 Hz will be called as 5^{th} Harmonic.

OVERTONE NUMBER: If all the resonant frequencies of a system are arranged in ascending order and given serial numbers beginning with zero then the number corresponding to each frequency is called its overtone number. For instance if resonant frequencies are 50 Hz, 100 Hz, 150 Hz etc then 50 Hz (Fundamental) is zeroth overtone, 100 Hz is 1^{st} overtone, 150 Hz is second overtone etc.

→ For pipe open at both ends natural frequencies are $\nu = nv/2L$, here Harmonic Number (H.N.) $= n$, Overtone Number (O.N.) $= n-1$.

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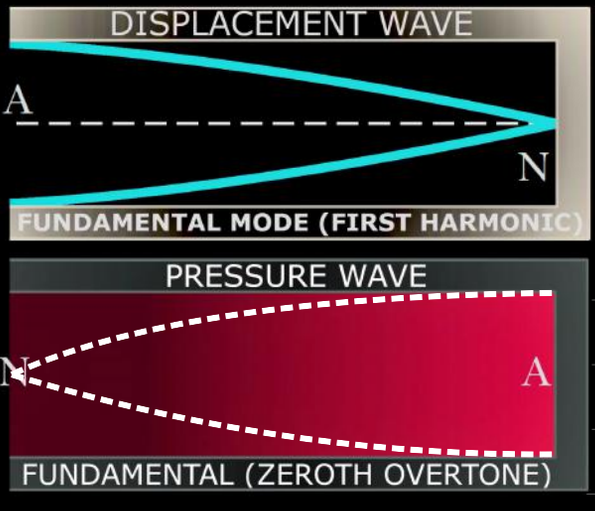
STATIONARY WAVES IN PIPE CLOSED AT ONE END:

FUNDAMENTAL MODE: Refer to the adjacent figure. The closed end is a displacement node while the open end is a displacement antinode. Here $L = \lambda/4 = v/4\nu$

$$\Rightarrow \boxed{\nu = v/4L}$$



FIRST OVERTONE (THIRD HARMONIC):

As clear from the adjacent figure,

$L = 3\lambda/4 = 3v/4\nu$

$\Rightarrow \quad \nu = 3v/4L$

→ Note that after $v/4L$ we have $3v/4L$ as the next resonant frequency thus only odd harmonics are natural frequencies and even harmonics are all missing.

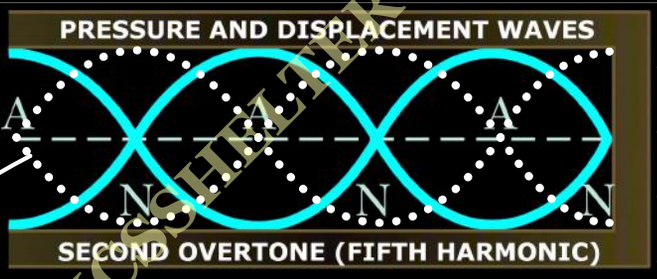
SECOND OVERTONE (FIFTH HARMONIC):

Refer to the adjacent figure, from the diagram we can see that :

$L = 5\lambda/4 = 5v/4\nu$

$\Rightarrow \quad \nu = 5v/4L$. Thus continuing the pattern, we get the complete set of natural frequencies:

$$\boxed{\nu = (2n+1)\frac{v}{4L}}$$



→ PRESSURE WAVE.

OVERTONE NUMBER (O.N.) = n

HARMONIC NUMBER (H.N.) = $2n+1$

→ When a wind instrument is played, not only fundamental mode is excited but also higher harmonics are excited, the quality of sound depends on relative amplitudes of higher harmonics and that's what distinguishes the quality of sound of one instrument from the other. The quality of sound is also called Timber. (Probably because sound quality of woodwind instruments depends on the wood material from which it is made).

→ Similar excitation of various harmonics also occurs in string instruments like guitar, violin, Sitar, Sarod, Saarangi and Taampura etc and the quality of sound depends on shape, size and material of the resonance cavity.

→ In reed instruments like Harmonium, Harmonica, Accordion, Concertina etc, the air column is excited in to vibrations using the vibrations of a reed (A brass strip fixed on one end and free to vibrate).

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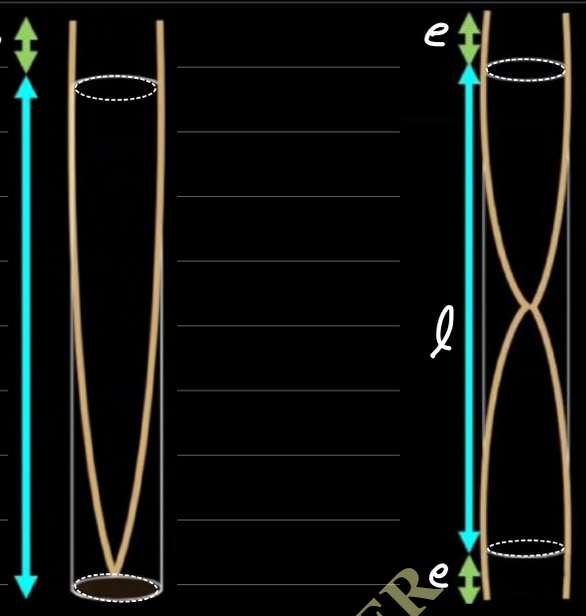
END CORRECTION:

So far we had assumed pressure node (or displacement anti-node) to be exactly at the open end, however in practice it is a little further. Thus for making more accurate calculations, we take the corrected effective lengths of the pipes.

For pipe closed at one end,

$$\boxed{l_{eff} = l + e}$$



For pipe open at both ends, $\boxed{l_{eff} = l + 2e}$ where $\boxed{e = 0.3D = 0.6r}$

D: Diameter of the pipe, r: radius of the pipe.

EXAMPLE: A closed organ pipe has length l and it is vibrating in 3rd overtone with maximum amplitude a. What is the amplitude at a distance $l/7$ from closed end.

Solution: Recall that first overtone has one full loop and another half loop, similarly second overtone has two full loops and another half, thus 3rd overtone has three full loops and another half. Thus:

$$3.5\,\frac{\lambda}{2} = l \Rightarrow \lambda = 4l/7 \Rightarrow k = 2\pi/\lambda = 7\pi/2l$$

Now closed end must be the node. Let the stationary wave be

$y = a\sin kx\cos\omega t = a\sin(7\pi x/2l)\cos\omega t$, Amplitude $= a\sin(7\pi x/2l)$

Thus amplitude at $l/7 = a\sin(7\pi/2l)\times(l/7) = a$ Ans.

EXAMPLE: Second overtone frequency of a closed pipe and fourth harmonic frequency of an open pipe are same. Find the ratio of 15th overtone of the closed pipe to the 12th overtone of open pipe.

Solution: Recall that for open pipe, $\nu = nv/2l$ where $n-1$ is the overtone number and n harmonic number. Also for closed pipe, $\nu = (2n+1)v/4L$ where n is overtone number and $(2n+1)$ harmonic number. Let l_c and l_o be lengths of closed and open pipes

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then according to the question, $(2\times2+1)\frac{v}{4l_c} = \frac{4v}{2l_o}$

$\Rightarrow \frac{l_o}{l_c} = \frac{8}{5}$. Now 15th overtone of closed pipe $= (2\times15+1)\frac{v}{4l_c} = \frac{31v}{4l_c}$

12th overtone of open pipe $(12+1)\frac{v}{2l_o} = \frac{13v}{2l_o}$, thus required answer, $(31v/4l_c)\,/\!/\,(13v/2l_o) = (31/26)(l_o/l_c) = \frac{31}{26}\times\frac{8}{5} = \frac{124}{65}$ Ans.

EXAMPLE: An open organ pipe with fundamental frequency 100 Hz and a closed pipe with fundamental frequency 150 Hz are joined in series. Find the resonant frequencies of new pipe.

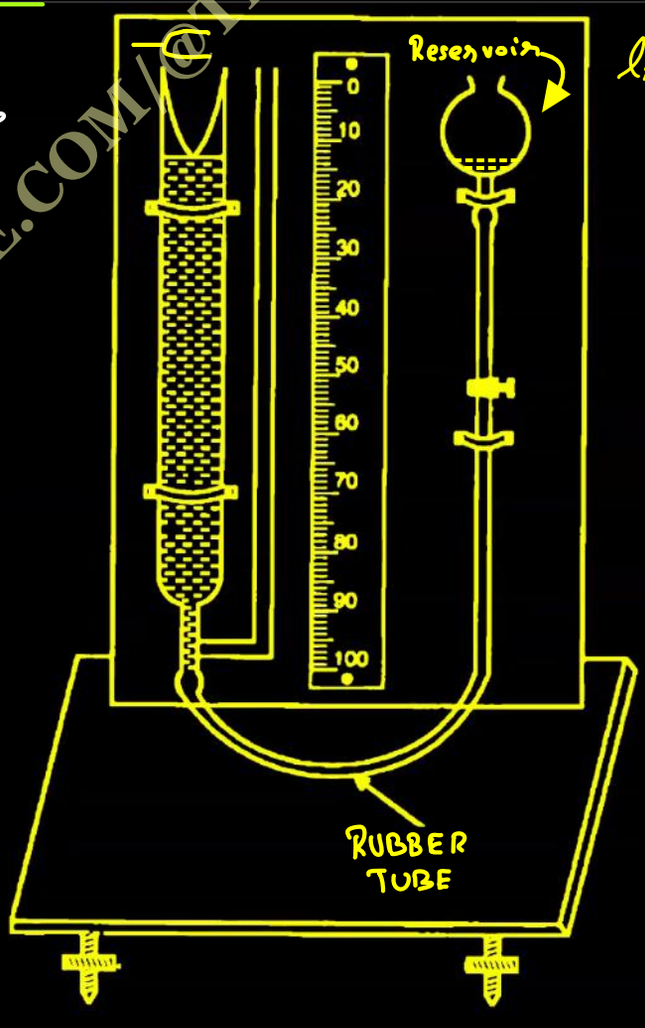
Solution: $\frac{v}{2l_o} = 100$ Hz $\Rightarrow l_o = v/200$ also $\frac{v}{4l_c} = 150 \Rightarrow l_c = v/600$

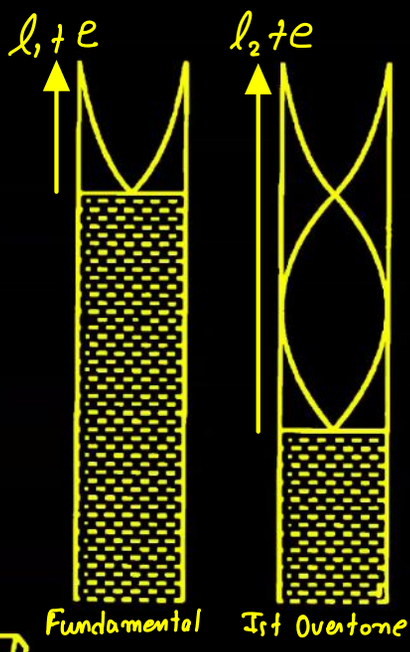
Required resonance frequencies $\nu = (2n+1)\,v/4(l_o+l_c) = \frac{(2n+1)\,v}{4(v/200 + v/600)}$

$\Rightarrow \nu = (2n+1)\times\frac{75}{2}$ Hz Ans.

RESONANCE COLUMN EXPERIMENT:

The adjacent figure shows the apparatus for resonance column experiment. The reservoir can be lowered or raised to adjust the level of water in the resonance tube which is excited using a tuning fork of known frequency. At resonance, a loud sound is observed and knowing the frequency of the fork and resonating lengths, end correction and speed of sound is calculated.





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We have for first resonance, $l_1 + e = \lambda/4 = v/4\nu \Rightarrow 4\nu(l_1+e) = v$ ①

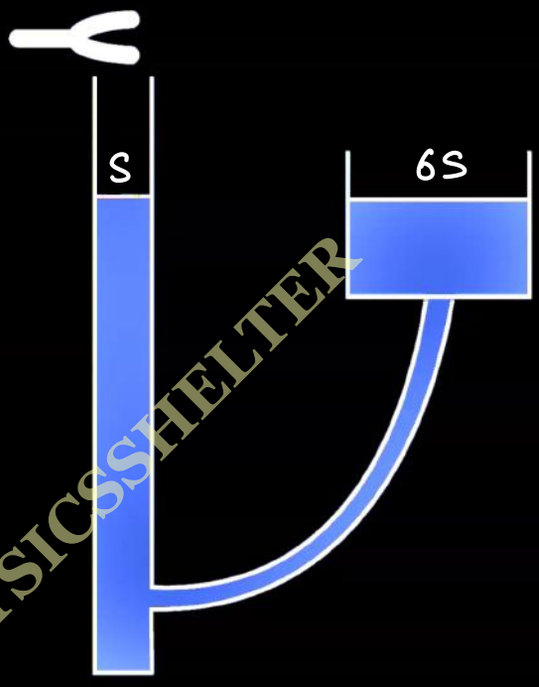
For second resonance, $(l_2+e) = 3\lambda/4 = 3v/4\nu \Rightarrow 4\nu(l_2+e) = 3v$ ②

Solving ① & ② we get:

$$\boxed{v = 2\nu(l_2 - l_1)} \qquad \boxed{e = (l_2 - 3l_1)/2}$$

EXAMPLE: SBT 69:

In resonance column experiment a tuning fork of frequency $f = 400\ Hz$ is held above the pipe as shown in figure. The reservoir is raised and lowered to change the level of water and thus the length of the column of air in the tube. The area of cross section of the reservoir is 6 times that of the pipe. Initially, the reservoir is kept so that the pipe is full up to the brim. Tuning fork is sounded and the reservoir is lowered. When the reservoir is lowered by 21 *cm*, first resonance is recorded. When the reservoir is lowered further by 49 *cm* the second resonance is heard. Find the speed of sound in air.



Solution: When reservoir is lowered by 21 cm, let water level in column fall by x cm then upper level of water in reservoir will also fall by x cm however base has fallen by 21 cm, thus volume entering the reservoir $= 6S(21-x)$ cm, this must be equal to volume reduced in pipe i.e Sx thus $Sx = 6S(21-x)$ $\Rightarrow x = 18$ cm. Similarly let second resonance length be y thus $Sy = 6S(21+49-y) \Rightarrow y = 60$ cm.

Now speed of sound, $v = 2\nu(l_2 - l_1) = 2 \times 400 (60-18) \times 10^{-2} = 336$

Thus speed of sound $= 336\ ms^{-1}$ Ans.

EXAMPLE: In a resonance tube experiment an 80 cm column of air is in resonance with a tuning fork in first overtone. Take downward direction as positive and $x = 0$ as the top point of tube. The speed of sound is $320\ ms^{-1}$. Write the possible equation representing family of possible standing waves. Max amplitude is A.

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Solution: $l = 3\lambda/4 = 80\,cm \Rightarrow \lambda = \frac{3.2}{3}\,m \Rightarrow k = 2\pi/\lambda = \frac{15\pi}{8}\,m^{-1}$

Now $\omega = v \times k = 320 \times 15\pi/8 = 600\pi\ rad\,s^{-1}$

Required equation family $A\cos(kx)\cos(\omega t + \phi) = A\cos\left(\frac{15\pi x}{8}\right)\cos(600\pi t + \phi)$ Ans.

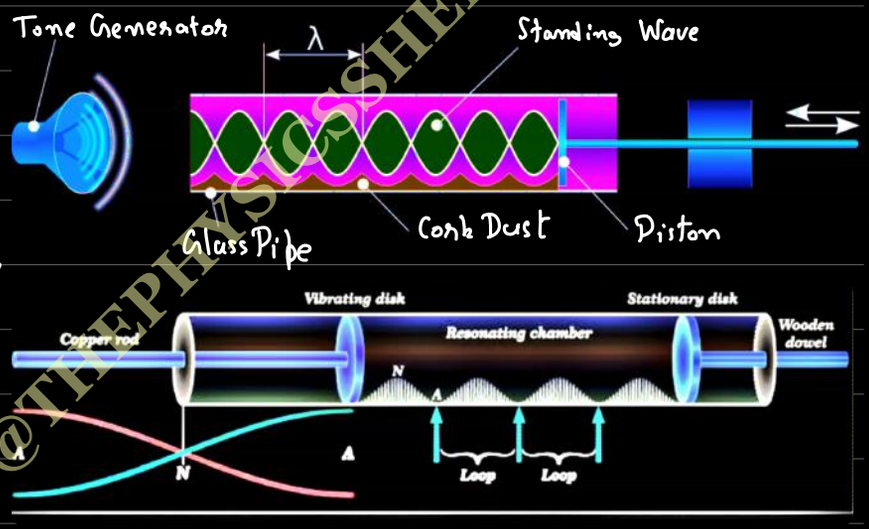
Think: In previous problem if max pressure amplitude were p_m what would be family of pressure wave equation?

Solution: $p = -B\,\partial y/\partial x = \frac{15\pi}{8} \cdot BA\sin\left(\frac{15\pi x}{8}\right)\cos(600\pi t + \phi)$

Thus required family of pressure waves $p = p_m \sin(15\pi x/8)\cos(600\pi t + \phi)$

KUNDT'S TUBE:

Kundt's tube is an apparatus for studying standing waves in air columns. It can be used to find velocity of sound, unknown source frequency or Young's Modulus of a metal rod by making suitable set ups. The lower diagram represents set up for finding Young's modulus of a copper rod.



The length of air column can be adjusted by setting the stationary disc to a desired spot. A copper rod clamped in middle is connected to a vibrating disc at its one end. Now copper rod is excited in its fundamental longitudinal mode of oscillation, thus exciting the air column in to standing waves. Saw dust in the air column is swept away from points of displacement antinodes, thus wavelength of sound in air is taken as twice the distance between two successive heaps of saw dust. Let this wavelength be λ_a. Now for resonance, frequency of metal rod must be same as frequency of air column. Thus:

$$\frac{v_m}{\lambda_m} = \frac{v_a}{\lambda_a} \quad (v_m \text{ velocity of wave in metal})$$

If L is the length of metal rod, $\frac{\lambda_m}{2} = L \Rightarrow \lambda_m = 2L$

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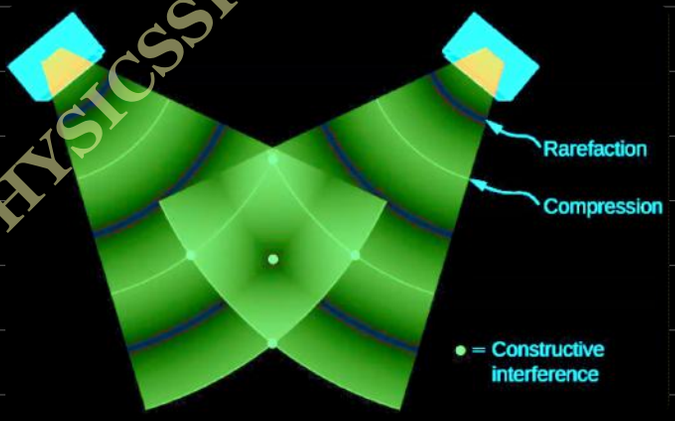
Thus: $\frac{1}{2L}\sqrt{\frac{Y}{\rho_m}} = \frac{v_a}{\lambda_a} \Rightarrow \boxed{Y = 4L^2\left(\frac{v_a}{\lambda_a}\right)^2 \times \rho_m}$

Here ρ_m is density of metal.

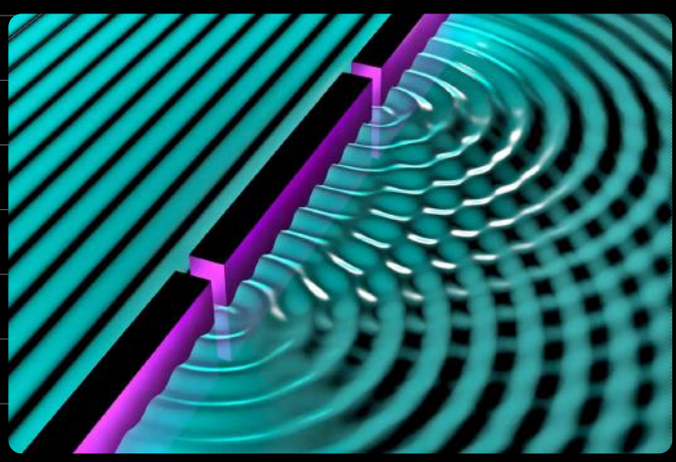
Alternatively if we know the Young's modulus of rod and its density we can rearrange this equation to solve for velocity of sound in air. We could use the setup of the first figure for finding the unknown frequency of tone generator (We can adjust the piston location until well defined nodes are formed (at this instant a loud sound is heard)). The frequency of tone generator is simply $\nu = v_a/\lambda_a$ where λ_a is simply double the distance between two successive heaps.

Interference Of Sound Waves:



Just like string waves, longitudinal waves also follow the superposition principle. Suppose p_1 is excess pressure at some point of space due to a wave ψ_1 and p_2 is excess pressure due to another wave ψ_2 then if both waves are simultaneously present, then net excess pressure will be simply $p_1 + p_2$. If phase difference between ψ_1 and ψ_2 is $2n\pi$ at some point of space we call it perfectly constructive interference and it leads to oscillations in large amplitudes at such points. If the phase difference is $(2n+1)\pi$ we call it destructive interference.

Coherent Sources: Sources oscillating with same frequency while maintaining same phase difference between them are called coherent sources. When waves from coherent sources interfere, a consistent interference pattern is obtained. i.e particles have well defined amplitudes depending on their locations. This amplitude depends on phase difference between



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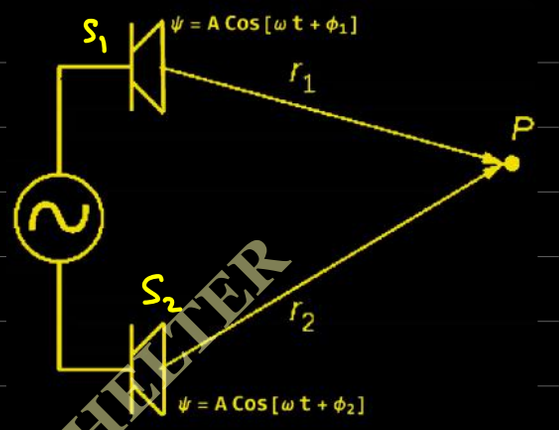
the waves from two sources. (For $2n\pi$ largest amplitudes are produced for $(2n+1)\pi$, (rest from one will coincide with trough from the other thus reducing the amplitude).

Calculating Phase Difference Between Waves At Any Location:

Consider two coherent sources of sound S_1 and S_2 producing single tone signals of angular frequency ω having phase angles ϕ_1 and ϕ_2 respectively. At certain point P the signals reach after covering distances r_1 and r_2 respectively, thus respectively time delays from source are respectively $\Delta t_1 = r_1/v$ and r_2/v. Now a time delay of T (one time period) causes a lag of 2π radians thus, lag produced in the two phasors due to travelling r_1 and r_2 distances are respectively $\theta_1 = \frac{\Delta t_1}{T} \times 2\pi = \frac{r_1}{vT} \times 2\pi = \frac{2\pi}{\lambda} r_1$



Similarly lag for lower phasor $\theta_2 = \frac{2\pi}{\lambda} r_2$. Now let respective phases by the time of reaching P be ϕ_2' and ϕ_1' then:

$$\phi_2' = \phi_2 - \theta_2 = \phi_2 - \frac{2\pi}{\lambda} r_2 \quad \text{Similarly,}$$

$$\phi_1' = \phi_1 - \theta_1 = \phi_1 - \frac{2\pi}{\lambda} r_1$$

Thus phase difference at $P = \phi_2' - \phi_1' = (\phi_2 - \phi_1) - \frac{2\pi}{\lambda}(r_2 - r_1)$

Or in short, $\boxed{\Delta\phi = \Delta\phi_0 - \frac{2\pi}{\lambda}(\Delta r)}$ where $\Delta\phi$ is initial phase difference $(\phi_2 - \phi_1)$ and Δr is path difference $(r_2 - r_1)$.

Interference Intensity: If I_1 and I_2 are individual intensities from the two sources at any point of space then intensity at the point of interference is given as $\boxed{I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\Delta\phi}$

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If $\Delta\phi = \pm(2n-1)\pi$ at some point we say n^{th} order minimum is formed at that point. For instance if $\Delta\phi = \pm\pi$ we call it first order minimum. Similarly if $\Delta\phi = 2n\pi$ at some point we say that n^{th} order maximum is formed at that point.

→ A minimum is also called a point of purely destructive interference and a maximum is also called a point of purely constructive interference.

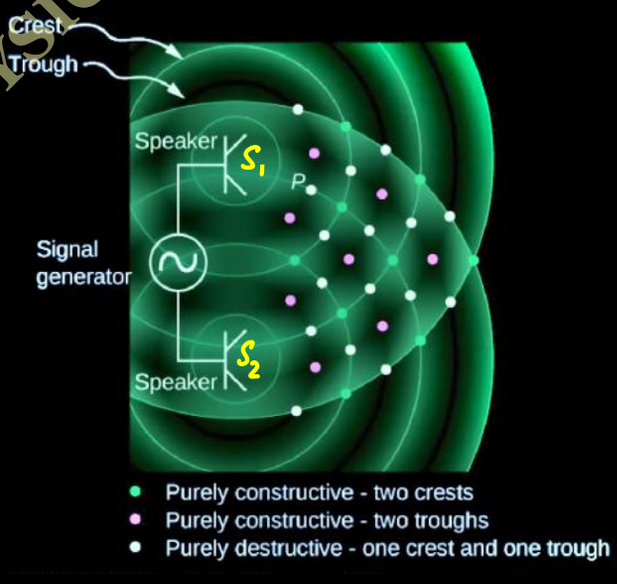
→ If $\Delta\phi_0 = 0$ then for a maximum, $\Delta x = n\lambda$ and for a minimum $\Delta x = (2n-1)\lambda/2$. This can be shown as follows:

$$\Delta\phi = \Delta\phi_0 - \frac{2\pi}{\lambda}\cdot\Delta x = \pm(2n-1)\pi \quad \text{(for minimum)}$$

putting $\Delta\phi_0 = 0 \Rightarrow \Delta x = (2n-1)\cdot\frac{\lambda}{2}$ similarly for constructive interference,

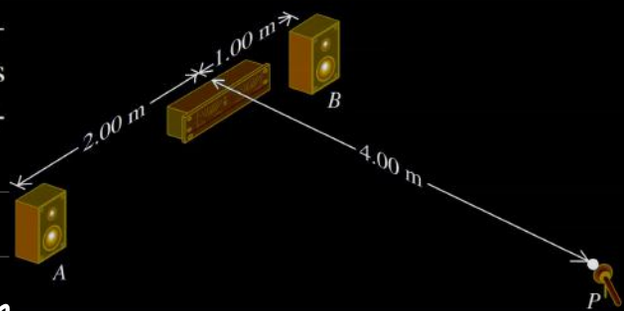
$$\frac{2\pi}{\lambda}\cdot\Delta x = 2n\pi \Rightarrow \Delta x = n\lambda. \quad \text{QED.}$$

→ Adjacent figure shows two speakers sounding in unison (same frequency) and $\Delta\phi_0 = 0$. The crest to trough spacing is $\lambda/2$ while crest to crest or trough to trough spacing is λ. Pure constructive interference occurs where a crest falls on a crest or a trough falls on a trough while destructive interference occurs where a crest falls on a trough. At all other points there is some intermediate phenomenon between purely constructive and purely destructive interference and intensity average is calculated there as $I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\Delta\phi$.



EXAMPLE:

Two small loudspeakers, A and B (**See figure**), are driven by the same amplifier and emit pure sinusoidal waves in phase. (a) For what frequencies does constructive interference occur at point P? (b) For what frequencies does destructive interference occur? The speed of sound is 350 m/s.



Solution: Path difference:

$$\Delta x = \sqrt{4^2+2^2} - \sqrt{4^2+1^2} = \sqrt{20} - \sqrt{17}\ \text{m}.$$

$$\Rightarrow \Delta x \approx 0.35\ \text{m}.$$

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Now for constructive interference, $\Delta x = n\lambda = nv/\nu \Rightarrow \nu = nv/\Delta x$

$\Rightarrow \nu = n \times 350/0.35 = 1000n$, thus required frequencies for constructive interference are 1000 Hz, 2000 Hz etc... (For 1000 Hz it will be first order maxima, for 2000 Hz it will be a second order maxima etc).

For destructive interference, $\Delta x = (2n-1)\lambda/2 = (2n-1)v/2\nu$

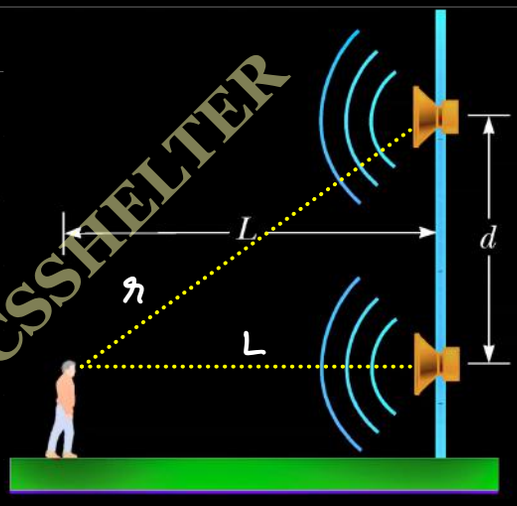
$\Rightarrow \nu = (2n-1)v/2\Delta x = (2n-1) \times 350/2(0.35) = (2n-1) \times 500$ Hz.

Thus for destructive interference, $\nu = 500$ Hz, 1500 Hz, 2500 Hz etc.

(For 500 Hz it will be a first order minima).

EXAMPLE:

Two speakers are driven by the same oscillator of frequency f. They are located a distance d from each other on a vertical pole. A man walks straight toward the lower speaker in a direction perpendicular to the pole as shown in Figure . (a) How many times will he hear a minimum in sound intensity? (b) How far is he from the pole at these moments? Let v represent the speed of sound and assume that the ground does not reflect sound. The man's ears are at the same level as the lower speaker.



Solution: Path difference $\Delta x = R - x = \sqrt{L^2+d^2} - L = (2n-1)\lambda/2$ (For min)

$\Rightarrow (2n-1)\frac{v}{2\nu} = \sqrt{L^2+d^2} - L$ (1), when $L \to \infty$ path difference goes to zero

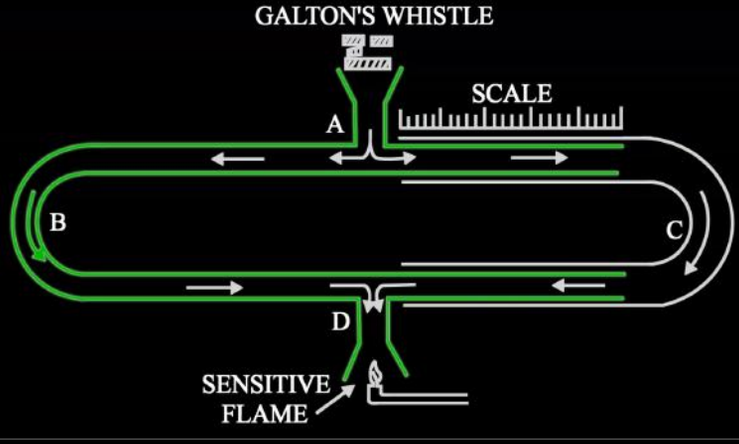
while when $L \to 0$ path difference becomes maximum of d. Now

$n = \frac{1}{2}\left(\frac{2\Delta x \nu}{v} + 1\right)$, thus n_{max} corresponds to Δx_{max} i.e. d

Thus $n = \text{GIF}\left[\frac{\nu d}{v} + \frac{1}{2}\right]$ Ans.

Now solving equation (1) for L, $L_n = \frac{d^2 - (n-1/2)^2(v/f)^2}{2(n-1/2)(v/f)}$ Ans.

QUINCKE'S TUBE: It is an apparatus which can be used for finding speed of sound in air. It consists of two U shaped sections that can slide one within the other. On one side there is a sound input arrangement for example Galton's Whistle (ultrasonic)



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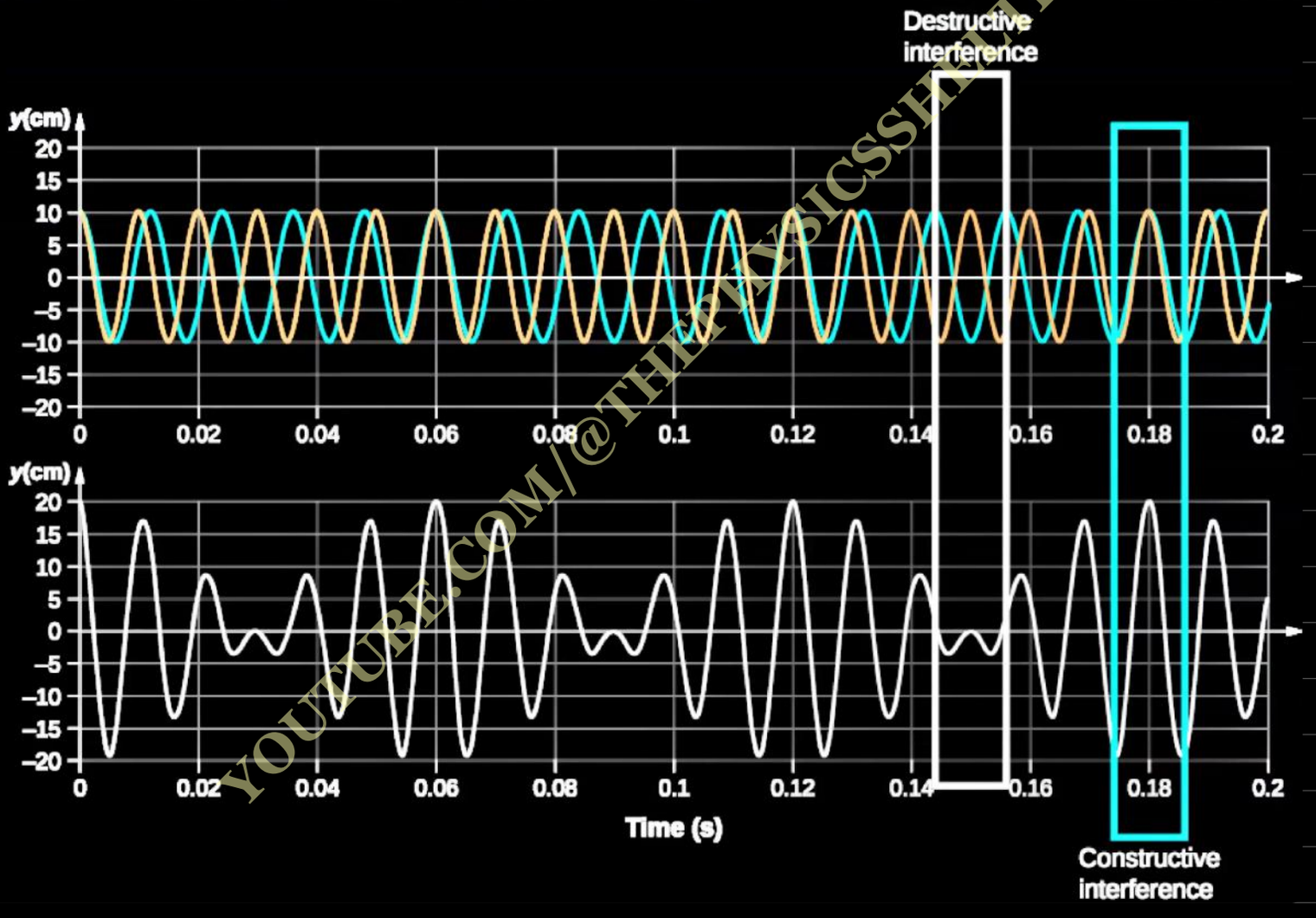
The sound reaches the flame through the two tube pathways ABD and ACD. If the path difference between the two paths is $n\lambda$, there is very intense sound (ultrasonic) causing flame to flicker violently. From this position if the inner tube is slid by a distance $\lambda/4$, path difference changes by $\lambda/2$ (i.e phase difference changes by π) and thus a minima is obtained. Thus we can find the wavelength of the sound and if the frequency of Galton whistle is known, velocity of sound is found simply using $v = \lambda\nu$.

BEATS: Consider the superposition of sound waves with slightly different frequencies



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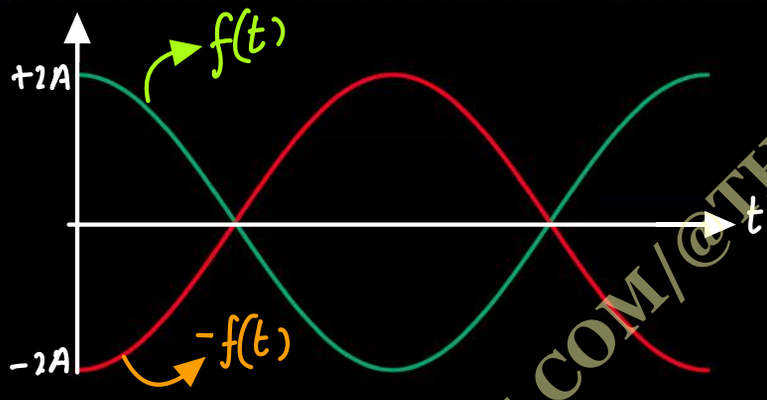
We see that due to slightly different frequencies, the phase difference between sounds keeps varying with time and accordingly we get sound of variable amplitude, Alternating becoming loud and feeble with continuously varying intensity. The number of times a loud sound is heard per unit time due to such a superposition is called beat frequency.

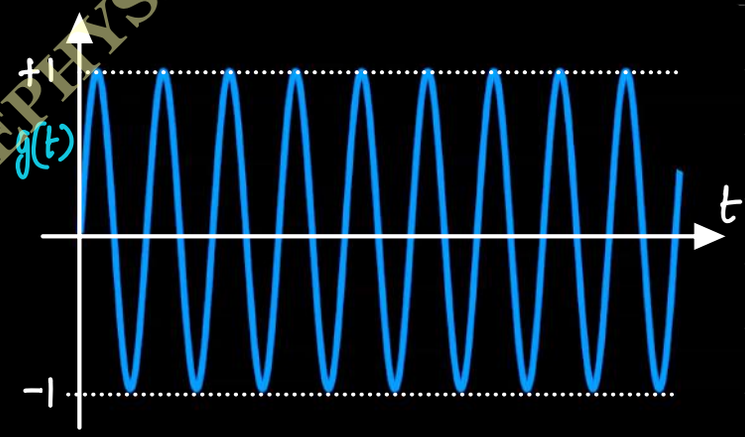
Now we will try to mathematically derive beat frequency produced by superposition of two sinusoidal waves. Consider particle displacement due to superposition of two sinusoidal wave, $y_1 = A \sin \omega_1 t$ $y_2 = A \sin \omega_2 t$. Then net particle displacement is given as:

$$y = y_1 + y_2 = A \sin \omega_1 t + A \sin \omega_2 t = 2A \sin\left(\frac{\omega_1+\omega_2}{2}\right) t \cos\left(\frac{\omega_1-\omega_2}{2}\right) t$$

Now this can be considered as product of two functions $f(t)$ and $g(t)$ where $f(t) = 2A \cos\left(\frac{\omega_1-\omega_2}{2}\right) t$ and $g(t) = \sin\left(\frac{\omega_1+\omega_2}{2}\right) t$.

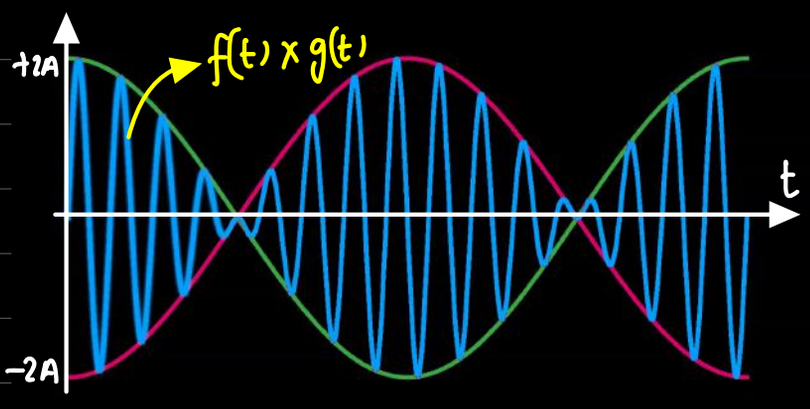
Recognise that $f(t)$ is a slowly varying function (Imagine $\omega_1 = 500$ and $\omega_2 = 501$ Rad/s) and $g(t)$ is a very rapidly varying function oscillating between ± 1. We shall try to individually visualise $f(t)$ and $g(t)$ before visualising the product.





$\because$ $g(t)$ oscillates between ± 1, $f(t) \times g(t)$ keeps oscillating between $f(t)$ and $-f(t)$.

A loud sound is heard whenever $f(t)$ is maximum or minimum. Thus beat frequency is double the frequency of $f(t)$. Now $f(t) = 2A \cos\left(\frac{\omega_1-\omega_2}{2}\right)$



$\Rightarrow$ Beat angular frequency :

$$\omega_{beat} = \left(\frac{\omega_1-\omega_2}{2}\right) \times 2$$

$$\Rightarrow \boxed{\omega_{beat} = |\omega_1 - \omega_2|}$$

Now $\nu = \omega/2\pi$, thus:

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$$\nu_{beat} = |\nu_1 - \nu_2|$$

Thus if we sound two tuning forks of frequencies 512 Hz and 516 Hz together, the beat frequency produced is 516 − 512 = 4 Hz.

Another Way Of Looking At Beats: Let $\omega_1 < \omega_2$ then relative to the phasor y_1, the phasor y_2 appears to rotate with $\omega_2 - \omega_1$. A loud sound is heard whenever two phasors align (and hence give a large resultant). Thus beat time period is given as:

$$T_{beat} = \frac{2\pi}{\omega_2 - \omega_1} \Rightarrow \nu_{beat} = \frac{\omega_2 - \omega_1}{2\pi} = \nu_2 - \nu_1$$

thus $\nu_{beat} = |\nu_2 - \nu_1|$.

→ If a tuning fork is filed its frequency increases
→ If a tuning fork is loaded with wax, its frequency decreases.

Example: A tuning fork with unknown frequency makes 3.00 beats per second with standard tuning fork 384 Hz. The beat frequency decreases when a small piece of wax is put on the prong of the first fork. What is the frequency of this fork?

Solution: $\nu_{beat} = |\nu_1 - \nu_2| = |\nu_1 - 384| = 3 \Rightarrow \nu_1$ could be possibly 381 or 387 Hz. Now loading with wax decreases the beat frequencies, thus its frequency must be on higher side i.e 387 Hz. (Because if its frequency were 381 Hz, its will further decrease and make the difference larger than 3 Hz).

Example: Two identical piano wires have a fundamental frequency of 600 Hz when kept under same tension. What fractional increase in tension of one wire will lead to a beat frequency of 6 Hz when both wires oscillate simultaneously?

Solution: $\nu = v/2\ell = \frac{1}{2\ell}\sqrt{\frac{T}{\mu}}$, thus $\frac{\Delta\nu}{\nu} = \frac{1}{2}\left(\frac{\Delta T}{T}\right)$

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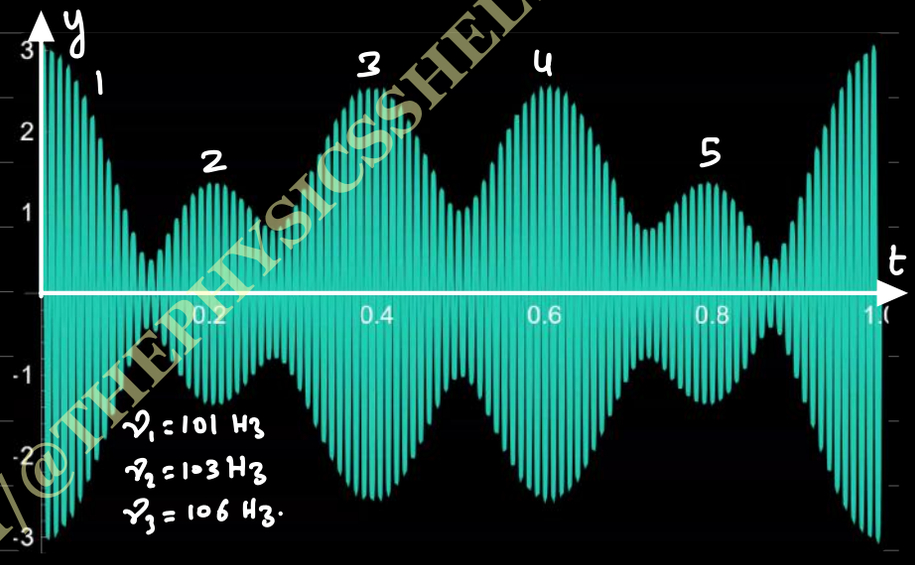
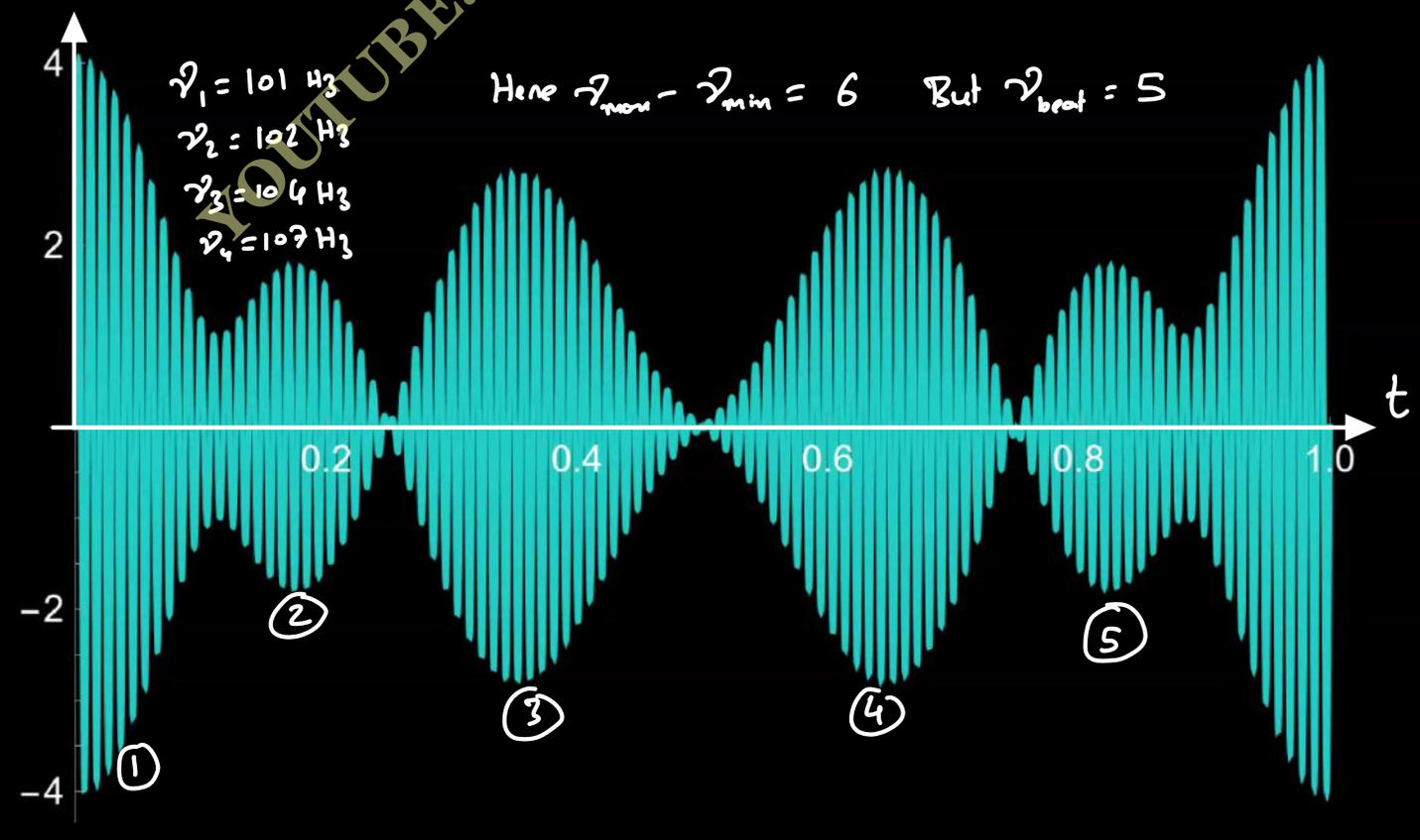
$$\Rightarrow \frac{\Delta T}{T} = 2\left(\frac{\Delta\nu}{\nu}\right) = 2 \times \frac{6}{600} = \frac{2}{100} = 2\% \text{ Ans.}$$

BEATS WITH THREE OR MORE FREQUENCIES: With three frequencies say ν_1, ν_2 and ν_3 with $\nu_3 > \nu_2 > \nu_1$, if the amplitudes are equal, beat frequency is simply the difference of highest and lowest frequencies i.e

$$\boxed{\nu_{beat} = \nu_3 - \nu_1}$$

with unequal amplitudes, beat frequency might be less than this, also for more than three frequencies with equal amplitudes, the difference of highest and lowest frequencies represents maximum possible beat frequency (actual frequency may be less than this number). As an example adjacent figure shows superposition of 101 Hz, 103 Hz and 106 Hz waves of equal amplitude (1 each) One can clearly observe 5 Beats (One primary and other secondary beats). Finding beats for a general case is complicated and requires plotting in general.





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Example: Two sources of frequencies 100 Hz and 102 Hz having same amplitude 'A' are interfering. A stationary detector can detect resultant amplitude greater than or equal to A. Find the fraction of time during which detector is active over a large amount of time.

Solution: Let the signals be $A\sin(200\pi t)$ and $A\sin(204\pi t)$

$\Rightarrow$ Net signal is $y = y_1 + y_2 = 2A\sin(202\pi t)\cos(2\pi t)$

thus amplitude is $2A|\cos(2\pi t)|$, now between $0°$ and $90°$, $\cos\theta$ is less than $1/2$ between $60°$ and $90°$ i.e $30°$ out of $90°$. Thus fraction of time amplitude is less than A is $1/3$. Thus fraction of time for which detector is active is $2/3$ Ans.

Example: Irodov 4.13:

A point A oscillates according to a certain harmonic law in the reference frame K' which in its turn performs harmonic oscillations relative to the reference frame K. Both oscillations occur along the same direction. When the K' frame oscillates at the frequency 20 or 24 Hz, the beat frequency of the point A in the K frame turns out to be equal to ν. At what frequency of oscillation of the frame K' will the beat frequency of the point A become equal to 2ν ?

Solution: From K frame, oscillations of particle are superposition of oscillations of K' frame and oscillations of particle relative to K' frame. The beat frequency representing motion of particle's oscillations must be the difference of the frequencies of K' and particle's oscillations in K' frame. Now since beat frequency is same at K' oscillating at 20 Hz or 24 Hz, The particle's frequency must be equidistant from these two frequencies. Thus from K' frame particle's must be oscillating at 22 Hz and beat frequency must be 2 Hz. Now to make the beat frequency 4 Hz, K' must oscillate at either 26 Hz or 18 Hz.

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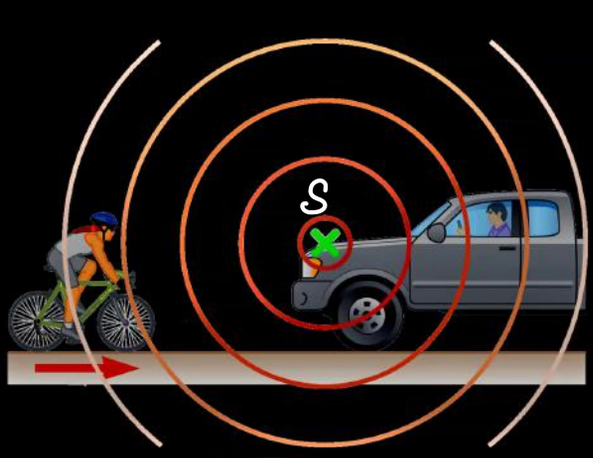
Doppler Effect: When either a source of sound is moving or the observer of the sound is moving (or both are moving) then the observed frequency of sound is different from emitted frequency. This phenomenon is called doppler effect.



This can be understood as follows:

If source is stationary but observer is moving towards source, he will be able to intercept the wavefronts more quickly as compared to the case if he were to simply wait for wavefronts to hit his ears.

If the source is moving, then it changes the wavelength in various directions, making the wavefronts denser in certain regions and farther apart in other regions, thus if a stationary observer is observing the sound, more or lesser number of wavefronts may hit his ears as compared to the case if the source were stationary.

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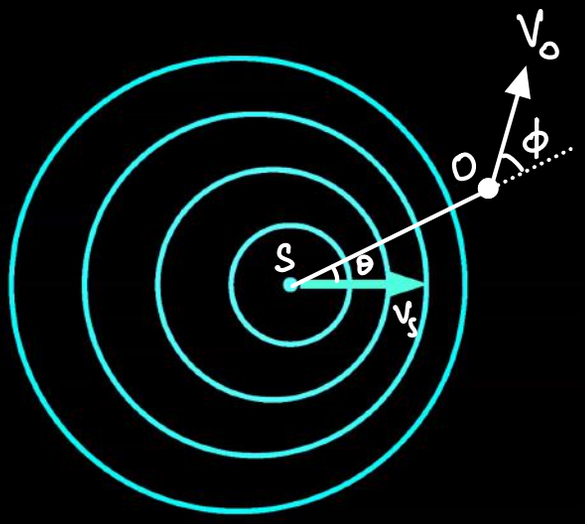
Derivation Of Observed Frequency In Doppler Effect

Let S be a source and O be an observer such that SO is the line joining the source to the observer (S is the position of the source when it emitted the sound (sometime in past) that is detected by observer now). Let $\vec{V}_S$ and $\vec{V}_O$ be the velocities of the source and the observer such that: $\vec{V}_S$ makes an angle with $\vec{SO}$ and $\vec{V}_O$ makes an angle ϕ with $\vec{SO}$, then we shall prove that:

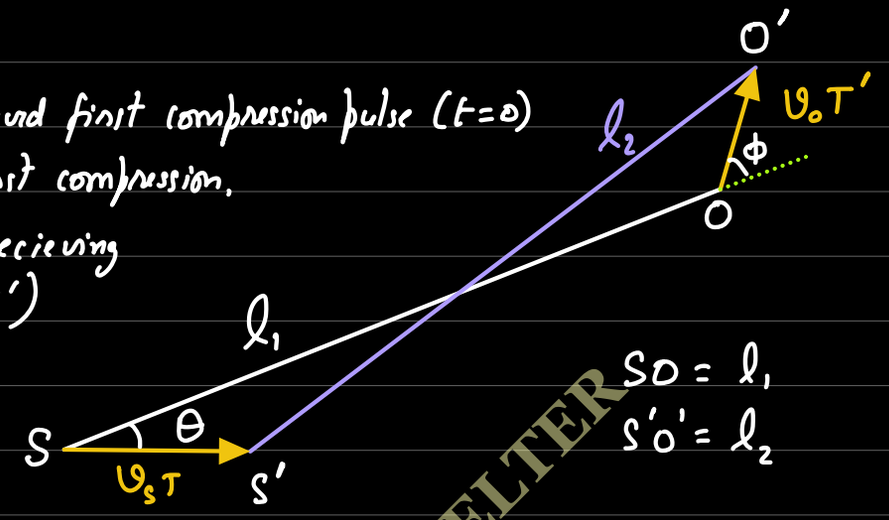


$$\nu' = \nu \left[\frac{c - V_O \cos\phi}{c - V_S \cos\theta} \right]$$

Where: ν → Frequency of the source
ν' → Frequency detected by observer
c → Velocity of sound.

PROOF:

O: Position of observer when it recieved first compression pulse ($t=0$)
S: Position of source when emitted first compression.
O': Position of observer at time of recieving second compression pulse ($t=T'$)
S': Position of source when it emitted second compression.
T: Time period between emissions.



$SO = l_1$
$S'O' = l_2$

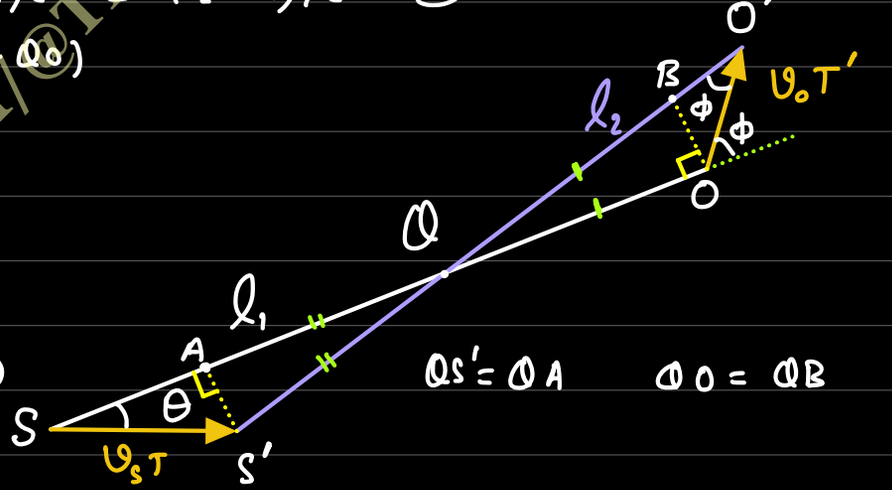
Now time coordinate of emission of first pulse $t_1 = -l_1/c$ ①
Time coordinate of emission of second pulse $t_2 = -l_1/c + T$ ②
Now time coordinate of recieving second pulse = time coordinate of emission + time taken for travelling, thus:

$$T' = -l_1/c + T + l_2/c = (l_2 - l_1)/c + T \quad ③$$

Now $l_2 - l_1 = (S'O + OO') - (SO + OO)$
$= (S'O - SO) + (OO' - OO)$
$= -v_sT\cos\theta + v_0T'\cos\phi$

Now using this in equation ③:

$$T' = (-v_sT\cos\theta + v_0T'\cos\phi)/c + T \quad ④$$



$QS' = QA$ $\quad QO = QB$

Now rearranging ④:

$$T'\left[1 - \frac{v_0\cos\phi}{c}\right] = T\left[1 - \frac{v_s\cos\theta}{c}\right] \Rightarrow T'[c - v_0\cos\phi] = T[c - v_s\cos\theta]$$

$$\Rightarrow \frac{1}{T'} = \frac{1}{T}\left[\frac{c - v_0\cos\phi}{c - v_s\cos\theta}\right] \Rightarrow \boxed{\nu' = \nu\left[\frac{c - v_0\cos\phi}{c - v_s\cos\theta}\right]} \quad (I)$$ QED

→ The formula (I) can also be written as:

$$\boxed{\nu' = \frac{c \pm v_{0N}}{c \mp v_{sN}}}$$

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Here subscript N represents component of velocity along the line joining the source and observer. The sign +/- is to be decided by common sense. Consider motion of one at a time (either source or observer). If due to its motion, the distance is decreasing, the frequency must increase (i.e + in numerator or − sign in denominator).

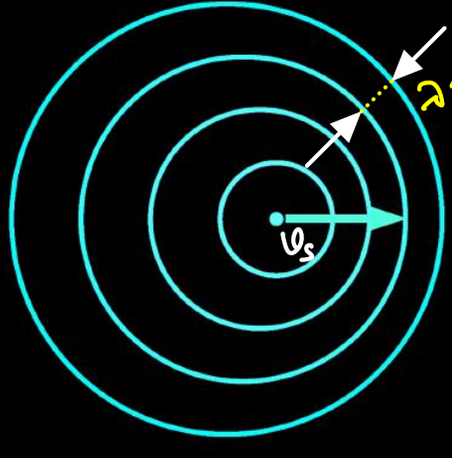
Doppler Effect With Collinear Velocities:

Doppler shift $f_o = f_s\left(\frac{v \pm v_o}{v \mp v_s}\right)$	Stationary observer	Observer moving towards source	Observer moving away from source
Stationary source	$f_o = f_s$	$f_o = f_s\left(\frac{v+v_o}{v}\right)$	$f_o = f_s\left(\frac{v-v_o}{v}\right)$ $c \to v$
Source moving towards observer	$f_o = f_s\left(\frac{v}{v-v_s}\right)$	$f_o = f_s\left(\frac{v+v_o}{v-v_s}\right)$	$f_o = f_s\left(\frac{v-v_o}{v-v_s}\right)$
Source moving away from observer	$f_o = f_s\left(\frac{v}{v+v_s}\right)$	$f_o = f_s\left(\frac{v+v_o}{v+v_s}\right)$	$f_o = f_s\left(\frac{v-v_o}{v+v_s}\right)$

When the velocity of source or observer is along the line joining the source and the observer, it is a special case of the general formula that we derived for Doppler effect, with θ being 0 or π and ϕ also being 0 or π. The cases are summarized in the above table.

Wavelength In Arbitrary Direction:

The spacing between two successive compressions or rarefactions at at point along the direction of wave propagation at that point is called wavelength at that point. Thus wavelength is independent of motion of the observer and only depends on source. For purpose of finding the wavelength we can assume a stationary observer and then find the



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wavelength using the formula $\lambda' \nu' = c$ thus:

$$\boxed{\lambda' = c/\nu'}$$ Assuming stationary observer,

Now from formula (I), $\nu' = \nu\left[\frac{c - v_0 \cos\phi}{c - v_s \cos\theta}\right]$, now if we assume stationary observer, $\nu' = \nu\left[\frac{c}{c - v_s \cos\theta}\right] \Rightarrow \lambda' = \frac{c}{\nu'}$

$$\Rightarrow \lambda' = \frac{c - v_s \cos\theta}{\nu} = \frac{c - v_s \cos\theta}{c/\lambda} = \lambda\left[1 - \frac{v_s}{c}\cos\theta\right]$$

thus $\boxed{\lambda' = \lambda\left[1 - \frac{v_s}{c}\cos\theta\right]}$ Here λ is wavelength is source were stationary.

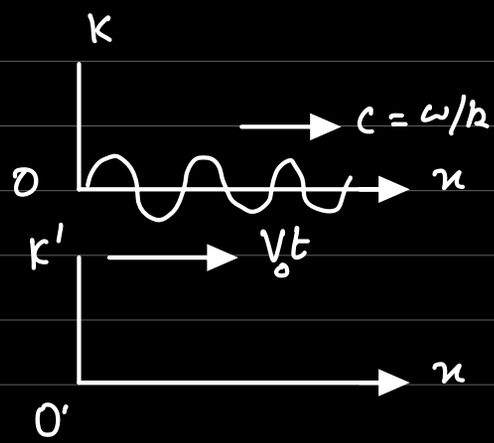
→ least wavelength occurs along the direction of motion of source i.e $\theta = 0°$. This least wavelength is $\lambda' = \lambda\left[1 - \frac{v_s}{c}\right]$. This is defined only when $v_s < c$.

→ Maximum wavelength occurs in a direction opposite to the motion of the source (i.e $\theta = 180°$), there $\lambda' = \lambda\left[1 + \frac{v_s}{c}\right]$.

EXAMPLE : IRODOV 4.153:

A plane elastic wave $\xi = a\cos(\omega t - kx)$ propagates in a medium K. Find the equation of this wave in a reference frame K' moving in the positive direction of the x axis with a constant velocity V_0 relative to the medium K. Investigate the expression observed.

Solution: Since K' is travelling at a speed V_0, events occuring at a coordinate x in K frame must be occuring at $x' = x - V_0 t$ in the K' frame.



Now:

$$\xi = a\cos(\omega t - kx) = a\cos(\omega t - k(x' + V_0 t))$$

$$\Rightarrow \xi = a\cos[(\omega - kV_0)t - kx'] \text{ Ans.}$$

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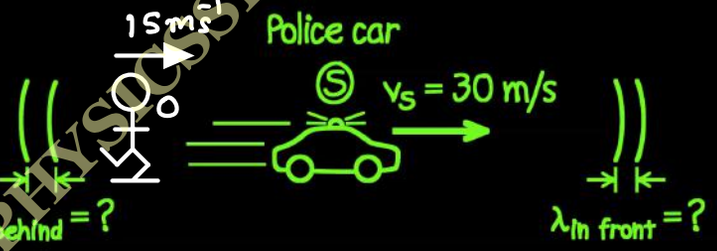
Note that frequency of this equation suggests that frequency of this wave in I' frame is $\omega' = \omega - kv_0 = \omega - \frac{\omega}{c} \times v_0 = \omega\left[1 - \frac{v_0}{c}\right]$

$\Rightarrow \nu' = \nu\left[1 - \frac{v_0}{c}\right]$, this is the same result as expected from doppler shift formula (Observer moving from away from source).

EXAMPLE :

A police car's siren emits a sinusoidal wave with frequency $f_S = 300$ Hz. The speed of sound is 340 m/s and the air is still. (a) Find the wavelength of the waves if the siren is at rest. (b) Find the wavelengths of the waves in front of and behind the siren if it is moving at 30 m/s.

Solution: $\lambda' = \lambda\left[1 - \frac{v_s \cos\theta}{c}\right]$



For car at rest, $\lambda = \frac{c}{\nu} = 340/300 = \frac{17}{15}$

$\Rightarrow \lambda = 1.13$ m Ans.

Now for front, $\theta = 0° \Rightarrow$

$$\lambda' = \frac{17}{15}\left[1 - \frac{30}{340}\right] = \frac{31}{30}\text{ m} = 1.033\text{ m Ans.}$$

For $\theta = \pi$, $\lambda' = \frac{17}{15}\left[1 + \frac{30}{340}\right] = \frac{37}{30}\text{ m} = 1.23\text{ m Ans.}$

Think Yourself: What is the frequency observed by the observer O in the previous situation?

Solution: $\nu' = \nu\left[\frac{c \pm v_0}{c \pm v_s}\right]$, here because of source movement distance is increasing thus decreasing the frequency, thus denominator must have a + sign. Now motion of observer alone would decrease the separation thus increasing the frequency, therefore numerator should also have a + sign.

Thus:

$$\nu' = \nu\left[\frac{c + v_0}{c + v_s}\right] = 300\left[\frac{340 + 15}{340 + 30}\right] \approx 288\text{ Hz Ans.}$$

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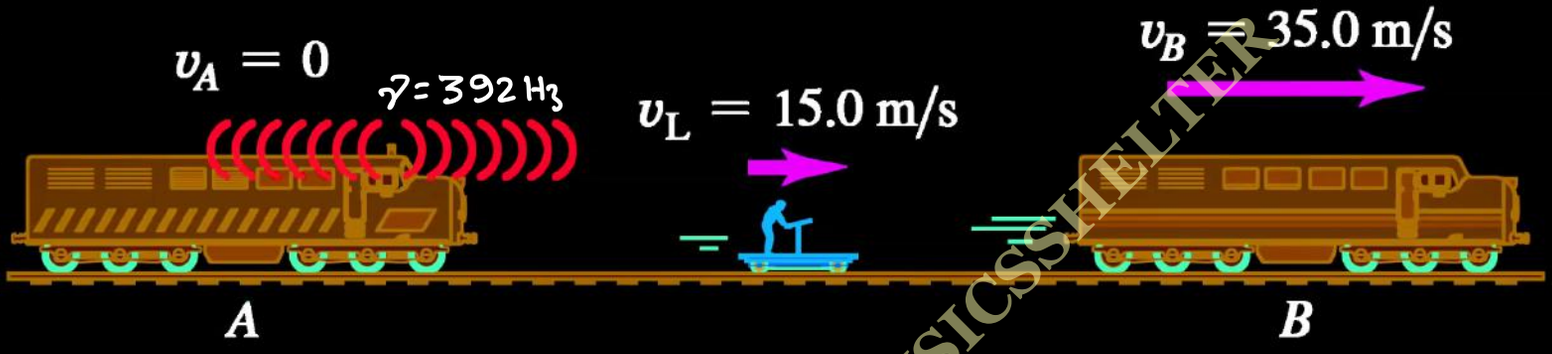
Doppler Effect With A Reflector (Double Doppler):

If a source is moving towards a reflector (could be moving or stationary). The reflector first acts as a moving observer and reflects doppler shifted waves, if the reflector is also moving, then it behaves like a moving source with source frequency equal to reflected wave frequency.

Example:



A stationary train 'A' sounds a horn of frequency of 392 Hz. The sound gets reflected from rear of train B and the listener L is hearing direct sound from A as well as reflected sound from B. Find the beat frequency at the listener's ears. Take $C = 330\ ms^{-1}$.

Solution: Frequency of direct sound from siren $\nu_A' = \left[\frac{330-15}{330}\right] \times 392$

$\Rightarrow$ Direct sound frequency at $L = 374.18\ Hz. = \nu_A'$

Direct sound frequency at reflector (B) $\nu_B' = \left[\frac{330-35}{330}\right] \times 392$

$\Rightarrow \quad \nu_B' = 350.42\ Hz.$ Now B becomes a source of ν_B', which is once again doppler shifted for the listener because both reflector and listener are moving. Now applying second doppler shift:

$$\nu_B'' = \nu_B' \left[\frac{330+15}{330+35}\right] = 331.22\ Hz.$$

Now beat frequency for the listener $\nu_{beat} = |\nu_A' - \nu_B''| = |374.18 - 331.22|$

$\Rightarrow \nu_{beat} \approx 43\ Hz$ Ans.

$\rightarrow$ Human ears can't percieve the beats if $\nu_{beat} > 10\ Hz$. It appears

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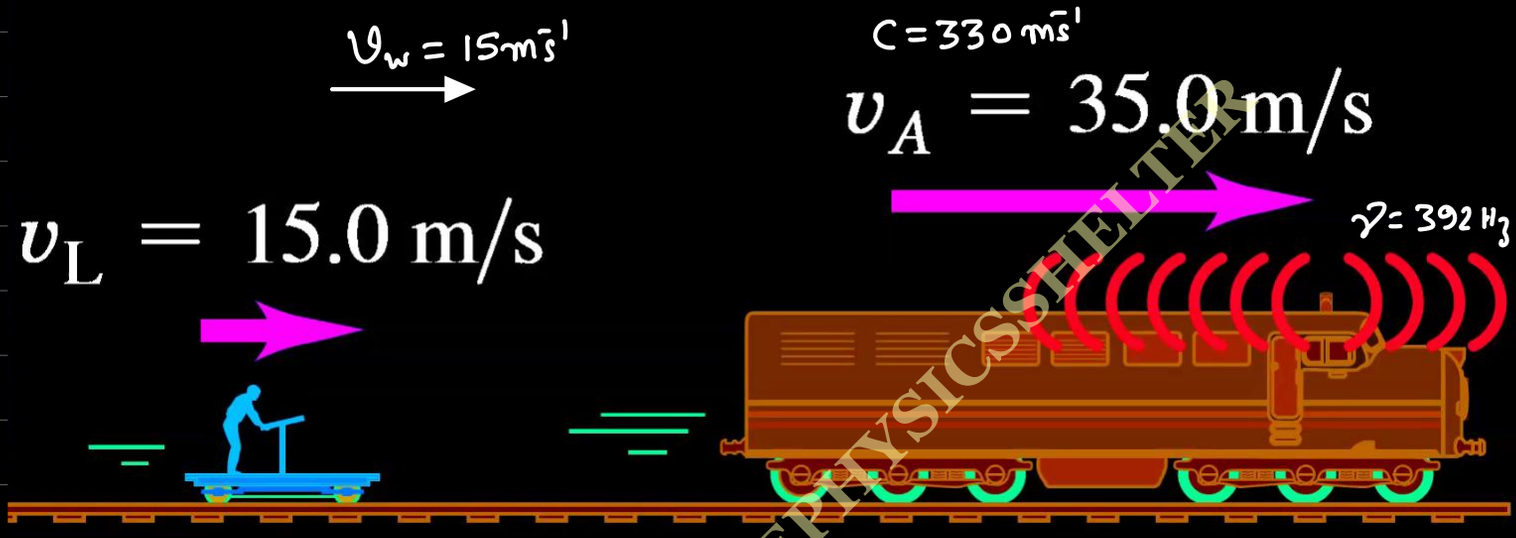
to be constant sound.

Doppler Effect In Presence Of Wind: If the medium of sound propagation is moving then we should convert the velocities of all emitters, reflectors and listeners to the frame of medium and after that we can apply standard doppler theory.

Example:



A listener is chasing a train as shown in the figure. The wind is also blowing at $v_w = 15 m s^{-1}$. Find the frequency observed by the listener.

Solution: Converting velocities from wind frame, $V_{Lw} = 0 m s^{-1}$, $V_{Aw} = 20 m s^{-1} \rightarrow$

Now applying doppler shift formula:

$$\nu' = 392 \left[\frac{330}{330+20}\right] \approx 370 \text{ Hz}. \quad \text{Ans.}$$

Doppler Effect With Oblique Approach: Recall the formula we had derived for general case of doppler effect:

$$\nu' = \nu \left[\frac{c \pm v_{0n}}{c \mp v_{sn}}\right]$$

with oblique approach, we just need to consider the components of V_0 and V_s along the line of approach. The line of approach is always defined as line joining the position of the source at the time of emitting the sound and location of listener at the time of listening to the sound. When the velocities of source or listener are sufficiently small, we may get approximately correct results

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even if we take positions of observer and source both at the time of listening (because due to small velocities, the change in angle of the line joining the source to observer will change very little during the transit time of the sound), however when velocity of source or observer are comparable to that of sound one must be very careful to choose the correct line of approach (line of approach is essentially the trajectory of the sound "ray" from source to listener).

Example: Irodov 4.178:

A source of sound with natural frequency $\nu_0 = 1.8$ kHz moves uniformly along a straight line separated from a stationary observer by a distance $l = 250$ m. The velocity of the source is equal to $\eta = 0.80$ fraction of the velocity of sound. Find :

(a) the frequency of sound received by the observer at the moment when the source gets closest to him;

(b) the distance between the source and the observer at the moment when the observer receives a frequency $\nu = \nu_0$.

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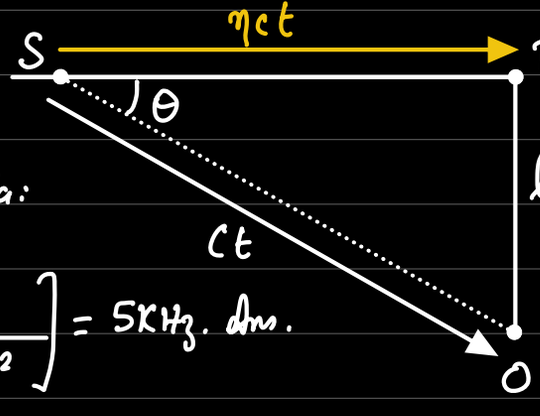
Solution: Here since velocity of source is $0.8c$, which is comparable to the velocity of sound, one must choose the line of approach very carefully. Let the sound that reaches the observer at the instant source is at P be emitted at a time t before. Then $SP = \eta ct$ and $SO = ct$

$\Rightarrow \cos\theta = \eta$. Now apply doppler shift formula:



$$\nu' = \nu\left[\frac{c}{c - \eta c\cos\theta}\right] = \nu\left[\frac{1}{1-\eta^2}\right] = 5\text{ kHz. Ans.}$$

(b) Recall $\nu' = \nu\left[\frac{c - v_0\cos\phi}{c - v_s\cos\theta}\right]$ (θ & ϕ from line $\vec{SO}$).

In our case $v_0 = 0$, thus if $\nu' = \nu$ then $\cos\theta = 0$, or the velocity of source must be $\perp$ to the line joining the source to the observer. This corresponds to the sound ray $s'o$. (See figure below).

Now $ct = l$, Now required distance:

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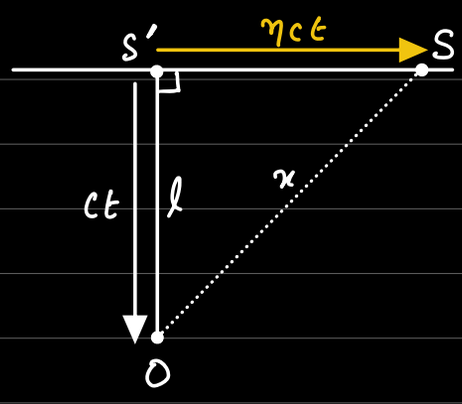
$$x = \sqrt{l^2 + (\eta ct)^2}$$

$$= l\sqrt{1+\eta^2} = 320\ m \quad \text{Ans.}$$



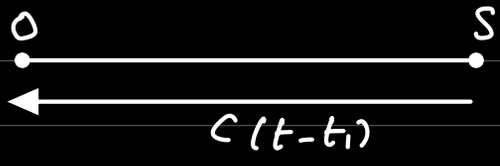
Doppler Effect With Accelerated Sources:

With accelerated source, the observed frequency will continuously vary with time. Here we will carefully need to find out velocity of the source at the time of emitting the sound using equations of kinematics. (Recall for accurately applying doppler effect equation, the source parameters are considered at the time of emitting the sound and listener parameters are considered at the time of recieving the sound.

Example: Irodov: 4.177

A source of sonic oscillations with frequency $\nu = 1700$ Hz and a receiver are located at the same point. At the moment $t = 0$ the source starts receding from the receiver with constant acceleration $a = 10.0\ m/s^2$. Assuming the velocity of sound to be equal to $c = 340\ m/s$, find the oscillation frequency registered by the stationary receiver $t = 10.0$ s after the start of motion.

Solution: Let the required sound ray corresponds to a time of emission t_1 after the beginning, thus time of travel of sound so $= c(t - t_1)$, since sound reached O at a time t. Now $OS = \frac{1}{2} a t_1^2 = c(t - t_1)$



Thus $t_1 = \frac{c}{a}\left[\sqrt{1 + \frac{2at}{c}} - 1\right]$, Now $v_s = at_1$

Thus required doppler shifted frequency:

$$\nu' = \nu\left[\frac{c}{c + v_s}\right] = \frac{\nu c}{c + c\left[\sqrt{1 + \frac{2at}{c}} - 1\right]}$$

$$\Rightarrow \nu' = \nu c / \sqrt{c^2 + 2atc} \quad \text{Ans.}$$

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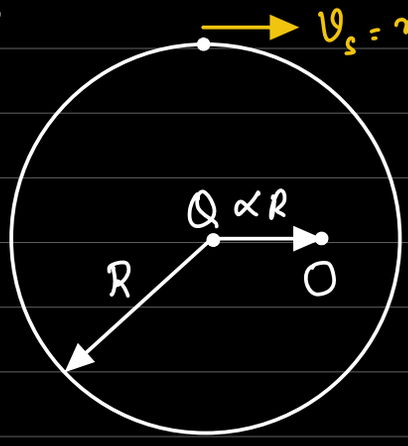
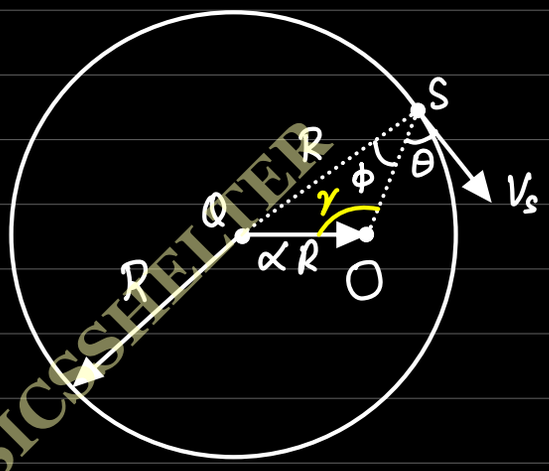
Example: A source is moving with a speed ηc along circumference of a circle of radius R. A point detector is placed inside the circle at a distance αR from its centre. Find the maximum and minimum possible frequencies recorded by the detector. Given source frequency ν.



Solution: For maximum observed frequency, the approach velocity of source towards listener must be maximum. Thus angle θ must be minimum. But $\theta + \phi = 90°$ thus we wish to maximize angle ϕ.

Now consider triangle QOS, $\frac{R}{\sin\gamma} = \frac{\alpha R}{\sin\phi}$



$\Rightarrow \sin\phi = \alpha \sin\gamma$, for max ϕ, $\sin\gamma = 1$

$\Rightarrow \sin\phi = \cos\theta = \alpha$.

Now max frequency: $\nu' = \nu\left[\frac{c}{c - \eta c\cos\theta}\right] = \nu\left[\frac{1}{1-\eta\alpha}\right]$

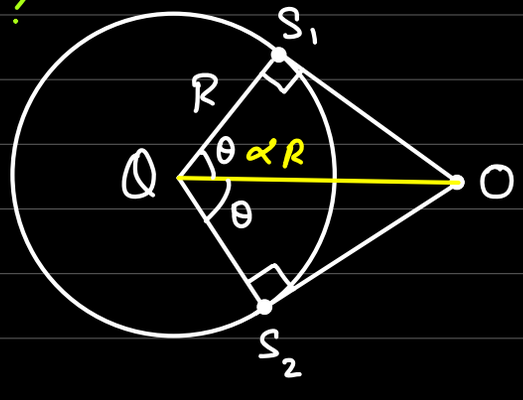
Thus $\nu_{max} = \frac{\nu}{1-\eta\alpha}$ Ans.

Similarly minimum frequency occurs when velocity of separation is maximum. Thus,

$$\nu_{min} = \frac{\nu}{1+\eta\alpha} \text{ Ans.}$$

Think Yourself: What would be your answer if $\alpha > 1$?

Solution: In this case head on approach / separation would be possible corresponding to sound rays S_1O and S_2O respectively.



Thus $\nu_{max} = \nu\left[\frac{c}{c-\eta c}\right] = \frac{\nu}{1-\eta}$ Ans.

$$\nu_{min} = \frac{\nu}{1+\eta} \text{ Ans}$$

→ By the time O hears maximum frequency, S_1 would have advanced through an angle $\delta = (\eta c/R) \times \left(\frac{S_1O}{c}\right)$

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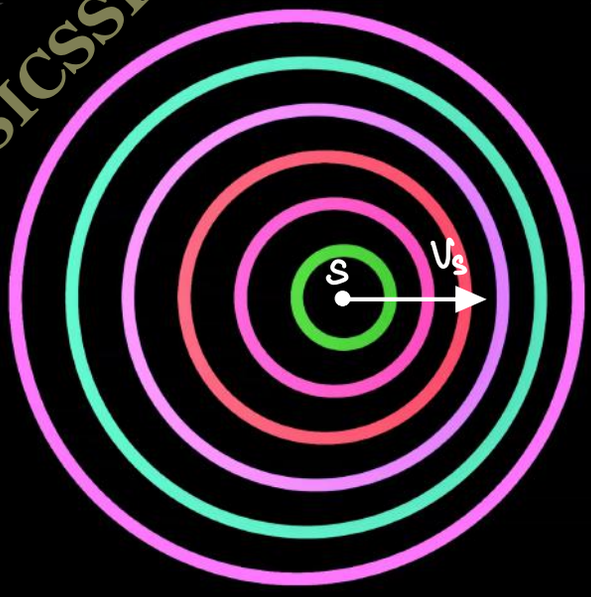
i.e $\delta = \frac{\eta c}{R} \cdot \frac{1}{c} \times \sqrt{(\alpha R)^2 - R^2} = \eta\sqrt{\alpha^2 - 1}$.

→ If source velocity is comparable to c be careful if position of source at the time of hearing maximum frequency is asked. (Remember to take the advance angle δ in to account).

→ Time to achieve minimum frequency after a maximum is in general not equal to the time required to achieve a maximum frequency starting from minimum. The ratios of these times depends on the ratio α. Thus from a frequency vs time graph it would be possible to figure out α using a little construction/geometry/trigonometry.

Energy Density Variation Due To Isotropic Moving Source:



If a source were stationary emitting isotropically an average Power P, then by the time the wavefront spreads to a radius R, the intensity would be $I = P/4\pi R^2$, however if the source is moving, the wavefronts in forward directions are squeezed together causing an increased energy density. For the same amount of total energy between two emitted wavefronts, energy density is inversely proportional to thickness of the distance between wavefronts. (Think of each wavefront emitting certain number of "acoustons" per unit area which is spread in a cylinder of thickness equal to local wavelength. Thus if 'u' were energy density at a wavefront of radius R, with moving source we can write energy density as $u' = u\frac{\lambda}{\lambda'}$. Now recall that we derived

$$\lambda' = \lambda\left[1 - \frac{v_s \cos\theta}{c}\right], \text{ thus } u' = u\left(\frac{c}{c - v_s\cos\theta}\right)$$

Similarly Intensity in arbitrary direction becomes $I' = I\frac{c}{c - v_s\cos\theta}$

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(∵ Now higher energy density is flowing with same speed through same area).

→ At sufficiently large distances from the source the wavefronts can be treated as planar in small volume. (Just like earth appears flat for small enough patches).

→ Recall for plane waves $u = \frac{1}{2}\rho A^2\omega^2$ and $I = \frac{1}{2}\rho c A^2\omega^2$, thus for isotropic source :

$$\boxed{u = I/c} \qquad \boxed{I = P/4\pi R^2}$$

P → Isotropic Source Power

R → Wavefront radius.

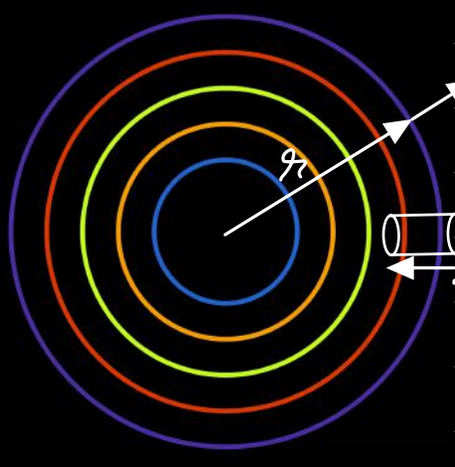
Alternatively, we can see, $I = P/4\pi R^2$, now in a time dt, energy travels a distance $dR = c\,dt$, This energy accumulates in a volume $4\pi R^2 dR$, thus, $u \times 4\pi R^2 dR = I \times 4\pi R^2 dt \Rightarrow u = I/c$.

UMOV VECTOR: A vector whose magnitude is equal to Intensity (i.e energy flow per unit time per unit area) and direction same as the wave velocity (or direction of energy flow) is called Umov Vector. (Essentially is is "Intensity Vector". $\vec{I} = I\hat{c} = u\vec{c}$. Umov Vector can be conveniently used to find Power flow (flux) through any area. $dP = \vec{I}\cdot d\vec{s}$.

EXAMPLE: PATHFINDER CHECK 13a: A point source is emitting power P isotropically. The velocity of sound is c. Now the source is stationary, and a detector moves with a constant speed v_0 towards the stationary source. Find suitable expression for intensity I of sound recieved by the detector at a distance r from the source.

Solution: Imagine a detector of cross section S moving head on towards the source. The velocity of detector in frame of local "acoustons" $= (c + v_0)$, thus volume swept by the detector in "acouston" frame $dV = S(c+v_0)dt$, Thus energy intercepted

$$dE = u\,dV = \left(\frac{P}{4\pi r^2 c}\right) S(c+v_0)dt \qquad (\because \text{Recall } u = I/c)$$



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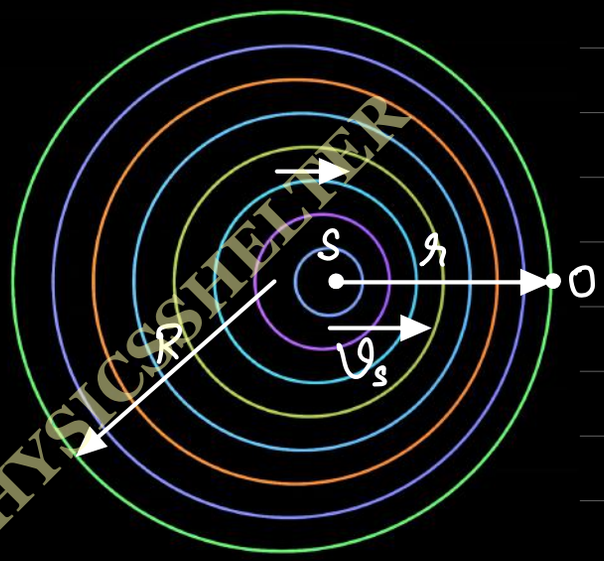
Thus $I = \frac{dE}{S dt} = \frac{P}{4\pi r^2}\left[1 + V_0/c\right]$ Ans.

Example: Pathfinder Check 13b: A point source is moving with a constant speed V_s towards a stationary detector in still air. Find suitable expression for Intensity I of sound recieved by the detector at a distance r from the source (r is the distance at the time of hearing the sound).

Solution: Recall that intensity in arbitrary direction is given by: $I' = I\left(\frac{c}{c - V_s \cos\theta}\right)$



where I and I' are intensities at a wavefront of radius R (with and without moving source). Now let observer be present at a wavfront of radius R as shown. Then intensity at O $I' = \frac{P}{4\pi R^2} \cdot \left(\frac{c}{c - V_s}\right)$

($\because \theta = 0$ & $I = P/4\pi R^2$). However we need to express this intensity in terms of distance between the source and observer (r) and not wavfront radius (R). Now let the wavfront be emenated t time ago then: $R = ct$ and $r = ct - V_s t \Rightarrow R = r\frac{c}{c - V_s}$

Thus $I' = \frac{P}{4\pi\left[r\left(c/c - V_s\right)\right]^2} \cdot \frac{c}{c - V_s} = \frac{P}{4\pi r^2}\left(\frac{c - V_s}{c}\right)$ Ans.

Example: Pathfinder Check 13c: A point source is moving in still air with a constant velocity V_s towards a detector which is moving with a constant velocity V_0 towards the source. Find suitable expression for intensity I recieved by the detector at a distance r from the source.

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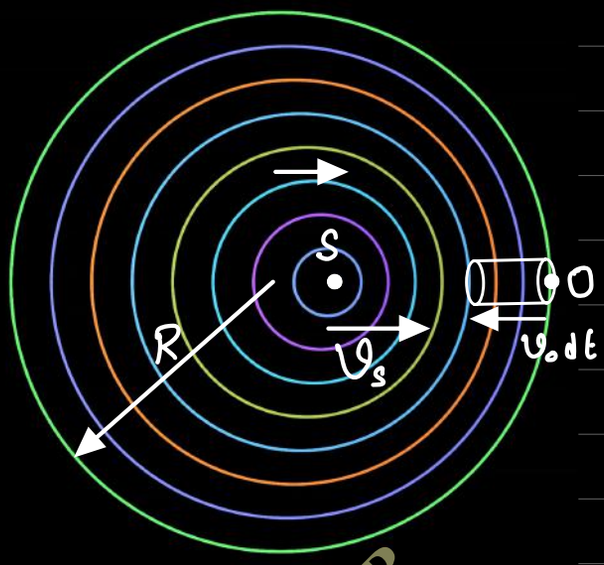
Solution: This is a combination of 13a and 13b (previous two examples). Now energy per unit volume, $u = I'/c$ (where I' is intensity calculated in previous example). If the detector has cross sectional area S, the volume swept by it in local acoustic's frame, $dV = (c+v_0)S\,dt$



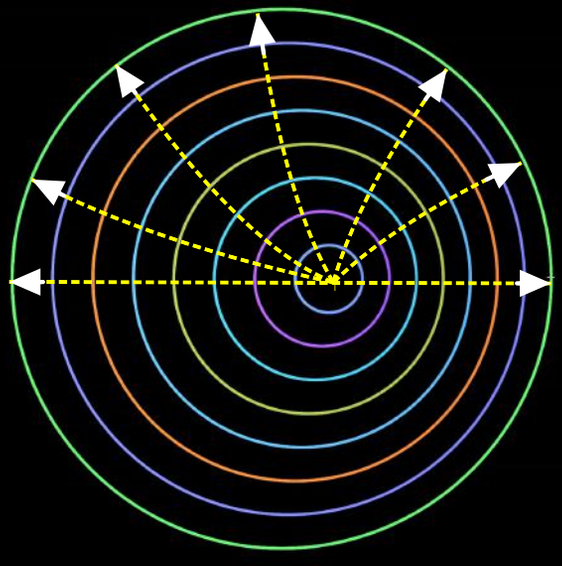
Thus $dE = u S (c+v_0) dt$

$\Rightarrow$ Required Intensity $I'' = \frac{dE}{S\,dt}$

$= u(c+v_0) = \frac{I'}{c}(c+v_0) = \frac{P}{4\pi r^2 c}\left[\frac{c-v_s}{c}\right][c+v_0]$

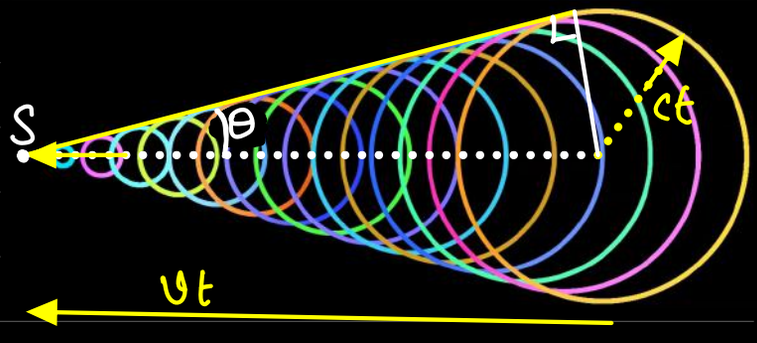
$\Rightarrow I'' = \frac{P}{4\pi r^2}\left[1+\frac{v_0}{c}\right]\left[1-\frac{v_s}{c}\right]$ Ans.

RELATION BETWEEN SOUND RAYS AND WAVEFRONTS: Sound rays represent the direction of energy travel at any location. The energy is transmitted along expanding wavefronts thus sound rays are always perpendicular to the wavfronts. For instance adjacent figure shows the wavfronts and some of the sound rays for a moving point isotropic source.



SHOCK WAVES (SONIC BOOM):

When a source is moving at a supersonic speed, it creates a conical surface of high pressure from its wavfronts released at differents points of time such that semivertex angle of the cone is given by



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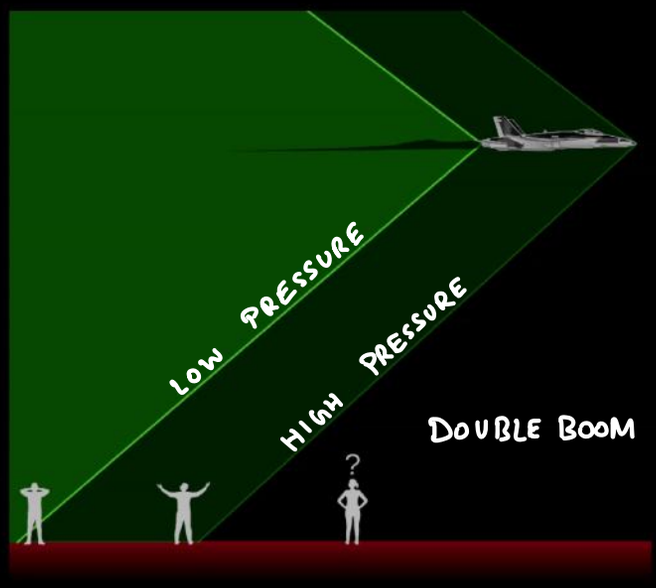
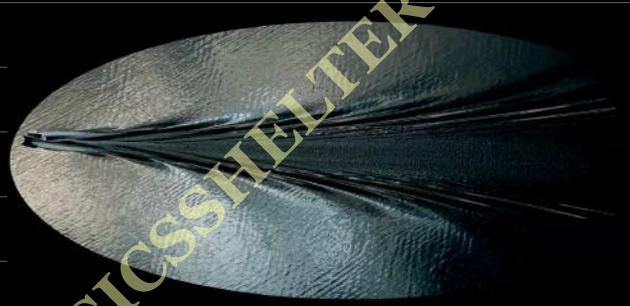
$\sin\theta = (ct/vt) = c/v.$

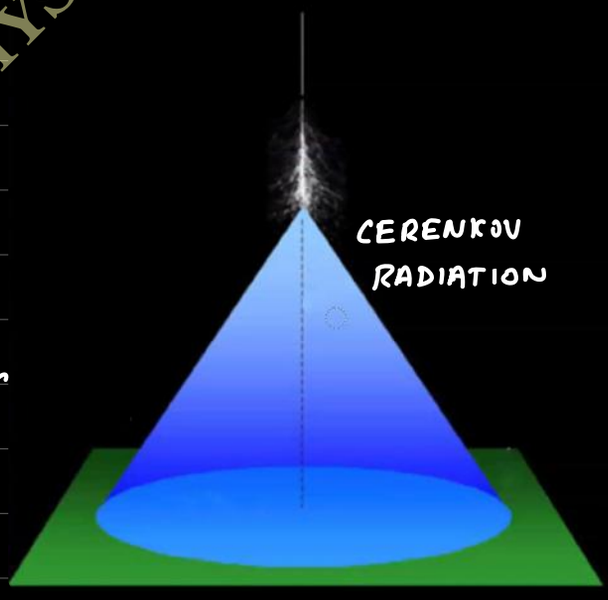
$$\sin\theta = c/v$$



When a supersonic aircraft is moving, the intersection of this conical surface with ground is a conic section (Hyperbola for horizontally moving aircraft). When this high pressure passes by an observer, the observer hears a very loud sound (known as sonic boom). A similar pattern is created by a boat on a swimming duck if boat velocity is greater than the speed of waves on surface of water.



A similar cone like formation is observed when a material particle such as an electron, positron or muon travels at a speed greater than the velocity of light in that material medium. When this happens, it is accompanied by emission of radiation known as Cherenkov radiation. Shock waves are also used in breaking up of kidney stones using a procedure called extracorporeal shock wave lithotripsy (Shock wave produced outside the body is focused on stone to break it).



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DOPPLER EFFECT FOR ELECTROMAGNETIC WAVES: If in the frame of the observer, a light source is receding with a velocity v directly away from the observer, the observed frequency is given as:

$$\nu' = \sqrt{\frac{1-\beta}{1+\beta}}\,\nu$$

where $\beta = v/c$

(For approaching source v is negative)

derivation of this result requires an understanding of special theory of relativity and we shall not derive it here. If $v \ll c$, we can simplify the formula using binomial approximation:

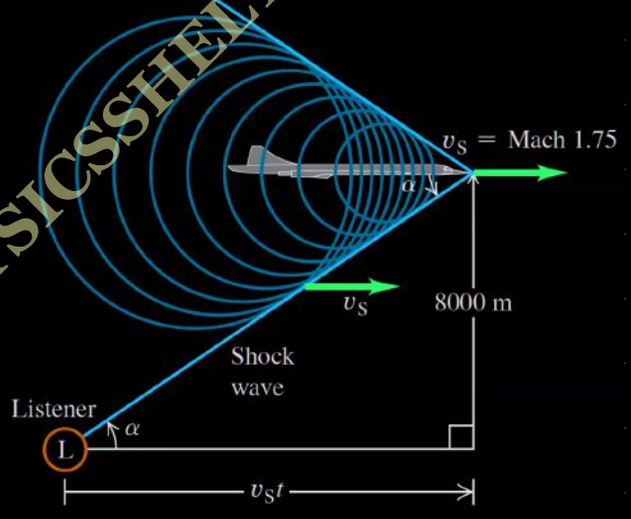
$$\nu' = \nu (1-\beta)^{1/2} (1+\beta)^{-1/2} \approx \nu (1-\beta/2)(1-\beta/2)$$

$$\Rightarrow \quad \nu' = \nu (1-\beta) \Rightarrow \frac{\nu - \nu'}{\nu} = \beta \Rightarrow \boxed{\left|\frac{\Delta\nu}{\nu}\right| = v/c}$$

→ Speed of light in vacuum is c in all frames ($c = 3\times10^8\ ms^{-1}$)

→ $c = \lambda\nu$ also holds for electromagnetic waves

EXAMPLE: An aeroplane is flying at Mach 1.75 at an altitude of 8000 m, where the speed of sound is 320 m/s. How long after the plane passes directly overhead will you hear the sonic boom?



Solution: We hear a sonic boom when shock waves reach our ears (or our ears are on surface of high pressure cone). Consider the figure above.

$\sin\alpha = c/v = 1/1.75 = 4/7$. Now $v_S t = 8000 \cot\alpha$

$$\Rightarrow t = \frac{8000\cot\alpha}{1.75c} = \frac{8000 \times (\sqrt{33}/4)}{1.75 \times 320} = 20.5 \text{ seconds. Ans}$$

EXAMPLE: Microwaves are reflected from a distant airplane approaching a stationary observer. It is found that when the reflected waves beat against the waves radiating from source, the beat frequency is 900 Hz. Calculate the approach speed of the airplane if wavelength of microwaves is 0.1 m. ($c = 3\times10^8\ ms^{-1}$).

Solution: First aeroplane acts as a moving observer and then acts as a moving source. Initial frequency $\nu_0 = c/\lambda$.

Now increased frequency observed at aircraft $\nu' = \nu_0(1+\beta)$

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Similarly when the waves reach the microwave source after reflection,

$$\nu'' = \nu'(1+\beta) = \nu_0(1+\beta)^2 = \nu_0(1+2\beta) \quad (\because \beta << 1)$$

$(\beta = v/c)$

$$\Rightarrow \frac{\nu'' - \nu_0}{\nu_0} = 2\beta \Rightarrow \frac{\nu_{beat}}{(c/\lambda)} = 2\beta \Rightarrow v = \frac{\nu_{beat} \times \lambda}{2c} \times c$$

$$\Rightarrow v = (\lambda \times \nu_{beat}/2) = 0.1 \times 900/2 = 45\ ms^{-1} \text{ Ans.}$$

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